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This Air Force Manual (AFMAN) implements Air Force Policy Directive (AFPD) 13-2, *Air Traffic Control, Airfield, Airspace and Range Management*. It applies to all Regular Air Force (RegAF) and including US Space Force until further notice, Air National Guard (ANG), and Air Force Reserve (AFR) organizations (to include contracted locations) who operate or administer functions and facilities for Radar, Airfield and Weather Systems (RAWS). At joint, shared-use, and overseas airfields, this manual applies to the facilities controlled and used exclusively by the Department of the Air Force, as outlined in international agreements, real estate documents or other written agreements. This manual outlines key duties and responsibilities of RAWS personnel. This manual also provides detailed guidance, procedures and programs for managing RAWS where the Air Force has functional oversight responsibility. This manual may be supplemented at any level; however, all Major Command (MAJCOM) supplements to include

interim changes to previously approved supplements must be routed to Air Force Flight Standards Agency, Director of Airfield Operations (AFFSA/XA) for coordination prior to certification and approval. References to MAJCOM within this publication refer to the MAJCOM Office of Primary Responsibility (OPR) for Airfield Operations. Unit (wing or base) level supplements to this publication must be routed to the responsible MAJCOM OPR for Airfield Operations for coordination prior to certification and approval. The authorities to waive wing/unit level requirements in this publication are identified with a Tier ("T-0, T-1, T-2, T-3") number following the compliance statement. See Air Force Instruction (AFI) 33-360, *Publications and Forms Management*, for a description of the authorities associated with the Tier numbers. Submit requests for waivers through the chain of command to the appropriate Tier waiver approval authority, or alternately, to the publication OPR for non-tiered compliance items or requirements. Refer to **Chapter 1** for additional waiver requirements to this manual. Refer recommended changes and questions about this publication to the OPR using the AF Form 847, *Recommendation for Change of Publication*; route AF Forms 847 from the field through the appropriate functional chain of command. Ensure all records created as a result of processes prescribed in this publication are maintained in accordance with AFI 33-322, *Records Management and Information Governance Program*, and disposed of in accordance with the Air Force Records Disposition Schedule located in the Air Force Records Information Management System.

(AETC) This supplement implements and extends the guidance contained in Air Force Manual (AFMAN) 13-204, Volume 4, *Radar, Airfield, and Weather Systems*. It applies to all Air Education and Training Command (AETC) units who manage airfield operations. This supplement contains guidance unique to the AETC training mission and further defines command requirements not specified in the basic AFMAN. This supplement does not apply to the Air National Guard, Air Force Reserve Command, or United States Space Force. This publication may be supplemented at any level, but all supplements must be routed to the OPR of this publication (19 AF/A3O) for coordination prior to certification and approval. Ensure that all records created as a result of processes prescribed in this publication are maintained in accordance with (IAW) AFI 33-322, *Records Management and Information Governance Program*, and disposed of IAW Air Force Records Information Management System Records Disposition Schedule. Refer recommended changes and questions about this publication to the 19 AF/A3OO Airfield Operations Branch using the DAF Form 847, *Recommendation for Change of Publication*; route DAF Forms 847 from the field through appropriate functional chain of command. The authorities to waive wing/unit level requirements in this publication are identified with a Tier ("T-0, T-1, T-2, T-3") number following the compliance statement. Submit requests for waivers through the chain of command to the appropriate waiver approval authority, through the publication OPR. Prior to initiating a **T-2** or higher waiver request requiring headquarters or higher coordination contact AETC Airfield Operations points of contact to discuss staffing actions required.

SUMMARY OF CHANGES

This document has been substantially revised and needs to be completely reviewed. Changes include but are not limited to: removal of non-essential acronyms and removal of numerous compliance terms/statements. In addition, tier waiver authorities as outlined in AFI 33-360 were validated and as necessary move to reside at the lowest appropriate level.

Chapter 1—GENERAL 7

1.1. Overview. 7
1.2. The Military and Civilian Aviation Integration Division. 7
1.3. Air Force Flight Standards Agency (AFFSA). 7
1.4. Operations at Contingency Locations. 7
1.5. Airfields Operated Pursuant to Contract. 8
1.6. Overseas Locations. 8
1.7. Waivers. 8
1.8. Governing Directives. 9
1.9. Issuing New Air Force Airfield Operations Policy or Procedures. 9
1.10. Maintenance Principals. 9

Chapter 2—ROLES AND RESPONSIBILITIES 11

2.1. AFFSA. 11
2.2. MAJCOM, FOA, or DRU. 11
2.3. Unit Commander. 12
2.4. Flight Commander/Director. 13
2.5. Flight Chief. 14
2.6. Maintenance Work Center Supervisors. 14
2.7. Regional Maintenance Center 19
2.8. System Managers will: 23

Chapter 3—MAINTENANCE WORK CENTER 24

3.1. Introduction. 24
3.2. Self-Assessment Program. 24
3.3. Mandatory Annual Reading List. 24
3.4. Air Traffic Control and Landing Systems Facility Requirements. 25
3.5. Air Traffic Control and Landing Systems Facility Records. 25
3.6. Air Traffic Control and Landing Systems Facility Reference Data. 26
3.7. Air Traffic Control and Landing Systems Maintenance Special Considerations for Removing Navigational Aid Identification Signals. 26
3.8. Other Equipment. 26
3.9. Work Center functions in support of Regionally Maintained Equipment. 27
3.10. ATC-Radio Antenna PMIs. 28

Chapter 4—MOBILITY WORK CENTER **29**

4.1. Introduction. 29

4.2. Chief of Maintenance. 29

4.3. Quality Assurance. 30

4.4. Maintenance Operations Center. 31

4.5. Materiel Control. 31

4.6. 1C8 Unit Type Code Team Lead. 32

4.7. Inspections and Evaluations. 33

Chapter 5—REGIONAL MAINTENANCE CENTER **35**

5.1. Introduction. 35

5.2. RMC Locations. 35

5.3. RMC Responsibilities. 35

Chapter 6—SYSTEM MANAGERS **36**

6.1. Introduction. 36

6.2. System Manager Functions. 36

6.3. System Manager Points of Contact. 37

Chapter 7—INSPECTIONS AND EVALUATIONS **38**

7.1. Introduction. 38

7.2. Inspection & Evaluation Functions. 38

7.3. Managerial Evaluations. 39

7.4. Personnel Evaluations. 39

7.5. Equipment Evaluation (EE). 42

7.6. Unit Type Code Evaluations. 44

Chapter 8—WORK CENTER SAFETY **46**

8.1. Work center safety functional areas. 46

8.2. Work Center Safety Program. 46

Chapter 9—TRAINING MANAGEMENT **48**

9.1. Introduction. 48

9.2. Training Program Requirements. 48

9.3. Climbing Training Requirements. 48

Chapter 10—LOGISTICS, LIFE CYCLE, AND PROJECT MANAGEMENT **49**

10.1. Introduction. 49

10.2. Air Force Centrally Supported Equipment/Systems. 49
10.3. Non-Centrally Supported Equipment/Systems. 49
10.4. Life Cycle Management Plans. 49
10.5. MAJCOM, Field Operating Agency, Direct Reporting Unit Functions. 50
10.6. Base/Unit 50
10.7. Work Center Supervisor. 51
10.8. Modification Management. 52

Chapter 11—MATERIEL MANAGEMENT 54

11.1. Introduction. 54
11.2. Logistics Readiness Squadron Liaison. 54
11.3. General Materiel Management. 55

Chapter 12—TECHNICAL ORDER MANAGEMENT 58

12.1. Introduction. 58
12.2. Work Center/Section/Unit Type Code Publication Management. 58
12.3. Technical Order Distribution and Control. 58
12.4. Local Work Cards. 58
12.5. Time Change Item Management. 59
12.6. Federal Aviation Administration Technical Manuals Management. 60
12.7. Air Force Maintenance Quality Control Checklist. 60
12.8. Air Force Maintenance Special Instruction (AFMSIs). 62

Chapter 13—TMDE AND TOOLS MANAGEMENT 64

13.1. Introduction. 64
13.2. General Test, Measurement and Diagnostic Equipment Requirements. 64
13.3. General Tool Requirements. 65
13.4. Work Center Tool Management. 65

Chapter 14—CORROSION PREVENTION AND CONTROL PROGRAM (CPCP) 68

14.1. Introduction. 68
14.2. Maintenance Work Center Functions. 68

Chapter 15—HISTORICAL RECORD MANAGEMENT 69

15.1. Introduction. 69
15.2. Overview. 69
15.3. Work Center Functions. 70

15.4. Completion of Air Force Technical Order Form 95. 73
15.5. Historical Records..... 74
Chapter 16—CONTRACT MAINTENANCE 76
16.1. Introduction..... 76
16.2. Compliance. 76
16.3. Contract Maintenance Management Responsibilities. 76
16.4. Contract Surveillance Programs/Contracting Office Representative..... 76
16.5. Contracting Officers. 76
16.6. Considerations. 77
Attachment 1—GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION 78
Attachment 2—EVALUATION REPORT WRITING AND SUBMISSION 89
Attachment 3—SAMPLING PLAN 94
Attachment 4—AIR FORCE MAINTENANCE QUALITY CONTROL CHECKLISTS 96
Attachment 5—AIR FORCE MAINTENANCE SPECIAL INSTRUCTIONS 97
Attachment 6—NEXT GENERATION RADAR OPERATIONS CENTER SUPPORT 98
Attachment 7—CRITICAL OUTAGE REPORTING 99

Chapter 1

GENERAL

1.1. Overview. This manual provides specialized and technical language and is intended for use by personnel who manage and/or maintain Air Force RAWS personnel, equipment, and facilities. This manual is complemented by AFMAN 13-204, Volume 1, *Management of Airfield Operations.*

1.2. The Military and Civilian Aviation Integration Division. The Military and Civilian Aviation Integration Division, Career Field Management is responsible for Air Force Specialty Codes (AFSCs) 13MX, Airfield Operations officer; 1C1X1, Air Traffic Control (ATC); and 1C7X1, Airfield Management (AM), 1C8X3, Radar, Airfield and Weather Systems (RAWS). Career Field Manager (CFM) duties will be executed in accordance with AFI 36-2101, *Classifying Military Personnel (Officer and Enlisted)*, AFI 36-2670, *Total Force Development*, and AFI 38-201, *Management of Manpower Requirements and Authorizations*. This list is not exclusive.

1.3. Air Force Flight Standards Agency (AFFSA). Air Force Flight Standards Agency is responsible for terminal area airfield operations (i.e., ATC, AM and RAWS) matters. In this capacity, AFFSA executes responsibilities as outlined in AFMD 27, *Air Force Flight Standards Agency (AFFSA)*, in keeping with policy guidance provided by the Air Staff. For the purpose of addressing specific procedural, training, standardization, architecture and integration issues directly related to airfield operations, AFFSA may have cause to interact with the MAJCOMs, Federal Aviation Administration (FAA), and other military services.

1.4. Operations at Contingency Locations. Contingency is defined as a situation requiring military operations in response to natural disasters, terrorists, subversives, or as otherwise directed by appropriate authority to protect United States interests. At contingency locations, outside the United States, the senior airfield authority (SAA) may waive airfield operations procedures outlined in this manual to support tactical or combat operations. This authority must not be exercised below the SAA level. **(T-1).** Prior to issuance of such a waiver, the following actions must be accomplished:

1.4.1. Conduct a risk management assessment with the rationale for the waiver and explain (1) how complying with the requirement/compliance item impacts mission accomplishment, OR (2) cost of compliance (training, funds, equipment, facilities, guidance or manpower) creates unacceptable risk to a higher priority task; OR (3) expected cost of compliance outweighs the benefit; OR (4) personnel cannot comply with the requirement due to a lack of resources (training, funds, equipment, facilities, guidance or manpower). **Note:** SAA waiver period must not exceed requested waiver period or 30 calendar days after the approving commander's deployment length, whichever is shorter. **(T-1).** Because waivers are the expression of a specific commander's risk acceptance, approved waivers automatically expire 30 calendar days after a change in commander, unless the new commander renews the waiver.

1.4.2. Forward proposed waivers to the Air Force Forces staff or equivalent for an operational review if time permits, or at the discretion of the SAA.

1.4.3. Refer to Air Force Tactics, Techniques, and Procedures (AFTTP) 3-4.4, *Contingency Airfield Operations* for additional information. This AFTTP outlines how Air Force Forces open and operate a deployed or contingency airfield. This AFTTP volume also contains

planning considerations for airfield management, ATC, and deployable air traffic control and landing systems capabilities, procedures, forces, and equipment.

1.5. Airfields Operated Pursuant to Contract. This manual applies to locations where DoD or Air Force access is pursuant to contract, to the extent specifically outlined in the contract statement of work (SOW) and/or performance work statement (PWS). Locations unable to comply with this criteria due to pre-existing SOWs are exempt until such time that the current contract expires. Locations that exclude this manual, shall specifically state rationale for the exclusion in the SOW. **(T-2).**

1.6. Overseas Locations. Any host nation procedures, or procedures based on otherwise inapplicable international regulations that are being implemented to fulfill basing or similar agreements should be implemented consistent with this manual to the extent possible. While agreed practices adopted for Air Force use in overseas areas take precedence, every effort should be made to conform to this manual.

1.6.1. MAJCOMs must identify any special procedures agreed with host nations in their supplement to this manual. Refer conflicting procedures to AFFSA/XA prior to implementation.

1.6.2. Airfield operations personnel augmenting a facility operated by another branch of the U.S. military will comply with their regulations and procedures as applicable. **(T-0).**

1.7. Waivers.

1.7.1. Units shall use the AF Form 679, *Air Force Publication Compliance Item Waiver Request/Approval* to process waivers to this manual. **(T-1).**

1.7.2. Unit waiver request must include the following:

1.7.2.1. A paragraph reference or text of the specific requirement for which the commander/director is requesting a waiver. **(T-1).**

1.7.2.2. Rationale for the waiver. **(T-1).** Explain which one of the following reasons apply and why: (1) How complying with the requirement/compliance item impacts mission accomplishment, (2) Cost of compliance (training, funds, equipment, facilities, guidance or manpower) creates unacceptable risk to a higher priority task; or (3) Expected cost of compliance outweighs the benefit; OR (4) personnel cannot comply with the requirement due to a lack of resources (training, funds, equipment, facilities, guidance or manpower).

1.7.2.3. Time period or circumstance the waiver is needed. **(T-1). Note:** Tier 1, 2, and 3 waivers may be approved for a period not to exceed the requested waiver period or 90 calendar after the approving commander's tour length, whichever is shorter. Because waiver are the expression of a specific commander's risk acceptance, approved waivers automatically expire 90 calendar days after a change in commander unless the new commander renews the waiver.

1.7.2.4. Risk mitigation measures to be implemented, if necessary, by the requesting commander during the waiver period. **(T-1)**.

1.7.2.5. Impact if waiver is disapproved. **(T-1)**.

1.7.3. If deemed necessary, submit additional data (e.g., letters of procedure, airspace maps, traffic patterns, airfield diagrams) to substantiate the waiver request.

1.7.4. Units must provide an informational copy of all Tier 3 approved waivers to MAJCOM OPR for airfield operations. **(T-2).**

1.7.5. Units will submit waivers to applicable FAA publications through their respective commander for coordination from the MAJCOM OPR for airfield operations, AFFSA/XA and the appropriate Air Force representative (AFREP). **(T-0).** The AFREP submits the request to FAA and notifies the requesting unit, MAJCOM OPR for airfield operations, and AFFSA/XA of the results.

1.7.6. The requestor's unit commander must forward a copy of the approved waiver to the AFFSA/XA workflow at **hqaffsa.xa@us.af.mil** within 30 calendar days of approval for situational awareness and process improvement considerations. **(T-1).**

1.7.7. Approved waivers become part of the approver's and requestor's official records and are appropriately filed in accordance with AFI 33-322. **(T-1).**

1.8. Governing Directives. Airfield operations services are governed by AFIs, AFMANs, and, to the extent applicable, FAA orders, parts of the Code of Federal Regulations, and any host nation procedures which have been adopted for Air Force use, unless specifically exempted or waived. Unit commanders may contact AFFSA through their respective MAJCOM OPR for assistance determining the applicability of directives.

1.9. Issuing New Air Force Airfield Operations Policy or Procedures. New policies or procedures are sent out via message and posted on the AFFSA Airfield Operations SharePoint® website. Airfield operations staff personnel are required to utilize the AFFSA Airfield Operations SharePoint® website to download and/or view messages.

1.10. Maintenance Principals.

1.10.1. The primary objective of maintenance is to ensure continued operational availability of systems and equipment used in support of the Air Force assigned operational mission. A key factor in achieving this objective is maintenance discipline. Maintenance discipline involves integrity and the employment of required skills in all aspects of the maintenance process. It is the responsibility of all personnel to comply with all written guidance and verbal instruction to ensure required repairs, inspections and documentation are completed in a safe, timely, and effective manner. **(T-1).** Managers should strive to establish a climate that promotes effective maintenance discipline.

1.10.1.1. **(Added-AETC)** Maintenance discipline also involves the continuous sharing of knowledge and team collaboration. Within AETC, there is a large contingent of civilian RAWS technicians. Managers must ensure the best use of civilian assets, encourage mentorship and training throughout the workcenter and between flights/sections. AETC RAWS operations will train and educate exceptional Airmen and exemplary technicians disciplined in as many RAWS systems as possible.

1.10.2. Technology is constantly changing, and traditional maintenance concepts must adapt to the new environment. The depth and breadth of these concepts vary considerably across the maintenance complex. Normally, older systems are more maintenance intensive than newer equipment sustained under the Line Replaceable Unit replacement concept. Personnel constantly endeavor to maintain their skills to adapt to these changes.

1.10.3. Readiness requirements may cause deferment of essential maintenance or training. Failure to recognize and support valid requirements may cause maintenance backlogs or mission failure. Managers will ensure every action is taken to make systems available for required maintenance actions and ensure maintenance is completed to meet mission requirements. **(T-2).**

Chapter 2

ROLES AND RESPONSIBILITIES

2.1. AFFSA. Establishes Air Force policies and guidance in specific 1C8 maintenance management, logistics and sustainment related areas. In pursuit of this, AFFSA will:

2.1.1. Develop RAWS maintenance management procedures and guidance.

2.1.2. Review waiver requests to this AFMAN and recommend appropriate actions.

2.1.3. Assess Air Traffic Control and Landing Systems (ATCALS) equipment as directed or requested.

2.1.4. Develop and review Self-Assessment Communicator checklists for this AFMAN.

2.1.5. Manage RAWS Air Force Maintenance Special Instructions (AFMSIs) and Air Force Maintenance Quality Control Checklists (AFMQCCs).

2.1.6. Provide guidance and support for RAWS maintenance, systems analysis, automated maintenance information systems, reliability, availability, and maintainability programs. Coordinate sustainment and logistics issues.

2.1.7. Provide geodetic survey support as it pertains to ATCALS.

2.1.8. Serve as Lead Command Manager for ATCALS. Work with the ATCALS Program Management Office (PMO) to resolve issues as they arise. Recommend priority of work from a safety of flight perspective and track workload to completion.

2.1.9. As requested, augment MAJCOM, FOA, or DRU Inspector General (IG).

2.2. MAJCOM, FOA, or DRU. MAJCOM, FOA, or DRU will implement the following guidance for their activities that perform RAWS maintenance functions. Regardless of size and type of organizational structure, the functions and duties outlined in this instruction are the minimum mandatory requirements necessary to ensure effective, quality maintenance is performed. The MAJCOM, FOA, and DRU representative will:

2.2.1. Forward validated waiver requests to this instruction to AFFSA/XA.

2.2.2. Concentrate management efforts on the unique needs within the command.

2.2.3. Establish MAJCOM, FOA, and DRU outage reporting instructions (e.g. when to utilize a Critical Outage Report (COR), which systems will require outage reporting, where to send the outage report). COR template is available in **Attachment 7** of this AFMAN.

2.2.4. Develop the minimum procedures necessary to accomplish the mission without generating unnecessary workload. Managers at all levels refine and improve these processes to minimize administrative, documentation, and reporting workload.

2.2.5. Serve as command focal point to implement Air Force guidance and directives concerning system modifications; programmed and mobile depot Technical Order (TO) improvements; source, maintenance, and recoverability code change requests; deficiency reports; product improvement reports; Management Internal Control Toolset (MICT) checklists; AFMSIs; corrosion control; maintenance management directives; and TO 00-series publications.

2.2.6. Provide life cycle logistics support for MAJCOM, FOA, and DRU -acquired Commercial Off-The-Shelf (COTS) and Government Off-The-Shelf (GOTS) systems, and equipment in accordance with **Chapter 10** of this AFMAN.

2.2.7. Manage the 1C8 portion of the Air Force Engineering and Technical Services (AFETS) program in conjunction with the MAJCOM, FOA, and DRU OPR.

2.2.8. Perform manpower, staffing, and developmental training actions as MAJCOM, FOA, and DRU -level RAWS functional management in accordance with Career Field Education and Training Plan (CFETP) and Air Force Qualification Training Package (AFQTP).

2.2.9. Obtain a Centralized Access for Data Exchange (CAFDEx)/Requirements Scheduling and Execution Program (RSEP) account to submit or validate Air Force Technical Order (AFTO) Form 227, *C-E Depot Maintenance Requirements and Schedule*. For further information on submitting AFTO Form 227, See TO 00-25-108, *Communications-Electronics (C-E) Depot Support*.

2.2.10. Identify a central Point Of Contact (POC) for administrative issues that will support the Regional Maintenance Center (RMC). Work closely with the RMC Superintendent and assigned units to resolve any policy and support issues adversely affecting the mission.

2.2.11. Evaluate and field-test proposed AFMQCCs prior to forwarding them to AFFSA/XA.

2.3. Unit Commander.

2.3.1. The unit commander or equivalent assigns responsibility for procedures outlined in this AFMAN. Only qualified personnel will accomplish inspection, preventive maintenance, servicing, opening, repair or replacement of RAWS equipment. **(T-1).**

2.3.2. The unit commander will:

2.3.2.1. Establish sound maintenance, supply discipline, and financial management practices in accordance with this AFMAN. **(T-3).**

2.3.2.2. Identify personnel required to maintain climbing proficiency and restrict climbing to structures necessary for mission accomplishment in accordance with AFMAN 91-203, *Air Force Occupational Safety, Fire, and Health Standards*. **(T-3).**

2.3.2.3. Fund safety related training, certification and equipment for RAWS maintenance. **(T-1).** Fund recurring expenses for certifications and equipment replacement. **(T-1).**

2.3.2.4. Fund in-residence training courses as necessary (e.g. equipment supplemental training, Quality Assurance (QA) course). **(T-3).**

2.3.2.5. Establish a Memorandum of Agreement (MOA) with the host base agency responsible for identifying and maintaining all mission critical circuits. **(T-3).** It is imperative that Defense Information Systems Agency scheduled interruptions, or loss of service be reported to the unit responsible for RAWS maintenance immediately. **(T-3).** This process will be identified in the MOA between the responsible agency and the squadron responsible for RAWS maintenance. **(T-3).**

2.3.2.6. Authorize and validate Basic Allowance for Subsistence (BAS) for RAWS personnel who are required to perform duties frequently at remote locations (off installation) or required to work unusual duty hours, where the use of filing DD Form 1475,

Basic Allowance for Subsistence- Certification, is impractical. **(T-3).** Commanders will validate BAS authorization in accordance with guidance established in DoD 7000.14-R, Volume 7A, *DoD Financial Management Regulation*. **(T-3).**

2.3.2.7. Appoint a work center Self-Inspection/Assessment Program Manager (SAPM), who will oversee the work center's self-inspection/assessment program as outlined in AFI 90-201, *The Air Force Inspection System*. **(T-3).**

2.3.2.8. Appoint a minimum of one RAWS technician to serve as the Inspections and Evaluations (I&E) evaluator, See **Chapter 7** of this AFMAN for specific details about the I&E role. **(T-3).**

2.3.2.9. Ensure logistics support is available for sustained operations for unit-acquired COTS/ GOTS systems/equipment. **(T-3).**

2.3.2.10. Approve/Disapprove the use of Local Maintenance Quality Control Checklists (LMQCC). **(T-3).** If approved forward to respective MAJCOM MFM/Functional Area Manager (FAM). **(T-3).**

2.3.2.11. **(Added-AETC)** Integrate all RAWS operations into the Airfield Operations Flight (i.e., OSM flight to OSAM section). **(T-2)**

2.4. Flight Commander/Director. Flight commanders/directors will: **Note:** If there is no flight commander/director, the Flight Chief will accomplish these duties. **(T-3).**

2.4.1. Advise and assist unit leadership, Flight Chief, work center supervisors and technicians in managing and administering maintenance programs. **(T-3).**

2.4.2. Control/manage personnel assignment and assigned duty positions within the maintenance activity and advise leadership on manning and skill level statuses. **(T-3).**

2.4.3. Participate in training status meetings at a minimum of every six months. **(T-3).**

2.4.4. Manage RAWS funding lines. **(T-3).**

2.4.5. Coordinate assistance with Subject Matter Experts (SME)/AFETS. **(T-3).**

2.4.5.1. SME/AFETS support is available to assist local RAWS technicians with a variety of maintenance actions and training.

2.4.5.2. Submit requests for assistance from these activities according to command directives. **(T-3).**

2.4.6. Verify installation project packages are reviewed for feasibility of installation and continued validity of the requirements. **(T-3).** This includes providing tools, test equipment, and support to work teams as required by the installation package or when needed to preclude work stoppages. **(T-3).**

2.4.7. Establish a Time Compliance Technical Order (TCTO) control program in accordance with TO 00-5-15, *Air Force Time Compliance Technical Order Process*. **(T-3).**

2.4.8. Establish a process for cannibalization requests in accordance with TO 00-20-2, *Maintenance Data Documentation*. **(T-3).**

2.4.9. Establish Local Work Cards (LWC) in accordance with TO 00-5-1, *Air Force Technical Order System*. **(T-3).**

2.4.10. **(Added-AETC)** Under the discrection of the AOF/CC integrate RAWS training, maintenance status, safety, and any desired program into standard AO meetings, such as AO's input to the Training Review Board, the Airfield Operations Board, and other locally established meetings. ATCALS inputs to Airfield Operations Boards shall include all system notifications and training issues impacting flight safety and the pilot production mission (for example, Training Visibility Ledger, discrepancies in task coverage, outages impacting mission accomplishment, etc.).

2.4.11. **(Added-AETC)** Ensure the most current civilian standardized civilian personnel core documents are used. **(T-3)**

2.4.12. **(Added-AETC)** Ensure training records are maintained by all 1C8 civilian personnel as required by their duties described in their SCPDs (9GK52 & 9GK51). **(T-3)**

2.5. Flight Chief. Flight Chief will: **Note:** If there is no Flight Chief, the maintenance work center supervisor will accomplish these duties. **(T-3).**

2.5.1. Verify an orientation program is established for newly assigned personnel. **(T-3).** The orientation program should augment rather than duplicate the unit orientation program. The program will provide a description of the mission(s); description and tour of maintenance staff and production work centers; description of individual duties and responsibilities; local policies (e.g., duty hours, leave, recall, safety, security); and other areas, as appropriate. **(T-3).**

2.5.2. Monitor TCTO actions (e.g., ensure TODAs review TO libraries assigned to them for TCTOs monthly, request extensions for any TCTO that will not be completed by rescission date, validate completion documentation). **(T-3).**

2.5.3. Track status of deferred Preventive Maintenance Inspection (PMIs). **(T-3).**

2.5.4. Approve or deny PMI cancellation requests. **(T-3).**

2.5.5. Ensure personnel complete official data calls and surveys soliciting feedback to improve current training or equipment sustainment issues and track tasking to completion. **(T-3).**

2.5.6. Verify work center supervisors have completed AFQTP 1C8X3-201D, *Work Center Managers Handbook*, within 6 months of assumption of duties. **(T-3).**

2.5.7. **(Added-AETC)** Inform AO leadership of any manning concerns and provide manning status updates quarterly. **(T-3)**

2.5.8. **(Added-AETC)** Inform AO leadership of any training gaps that cannot be filled locally and require outside agency support. **(T-3)**

2.6. Maintenance Work Center Supervisors. Work center supervisors will: **Note:** This section addresses the duties and responsibilities of individuals assigned to the position presiding over the RAWS maintenance work center. Duty titles may vary based on local organizational structure, reference 1C8X3 CFETP for approved duty titles.

2.6.1. Complete AFQTP 1C8X3-201D, within 6 months of assumption of duties. **(T-3).** The handbook can be found on the RAWS Training and Development website. Verify personnel have a thorough knowledge of their duties and comply with applicable directives and TOs. **(T-3).**

2.6.2. Verify compliance with maintenance schedules. **(T-3).**

2.6.3. Emphasize standard maintenance practices and enforce risk management to comply with safety standards. **(T-3). Note:** General maintenance practices can be found in TO 31-1-75*, General Maintenance Practices*.

2.6.4. Verify upgrade training and maintenance On-the-Job Training (OJT) programs are established and effectively managed. **(T-2).**

2.6.4.1. Complete duty position upgrade tasks in accordance with Master Training Plan (MTP). **(T-2).**

2.6.4.2. Correct observed or reported training deficiencies. **(T-2).**

2.6.4.3. Utilize all Air Education and Training Command (AETC) established training products and check the RAWS Training Development website for updates to applicable Air Force Job Qualification Standard (AFJQS)/AFQTPs. **(T-2).** This site is located at **https://usaf.dps.mil/teams/aetc-ksl-81trg/334/ULC/Documents/RAWS/1C8X3/RAWS%20Dashboard%26ToK/1. RAWS Dashboard.aspx?CT=1583760663124&OR=ItemsView**.

2.6.5. Maintain shift continuity via the following:

2.6.5.1. Review all open jobs for accuracy and to ensure the most up-to-date status/information is captured in IMDS or other approved AIS. **(T-3).**

2.6.5.2. Acquire appropriate Integrated Maintenance Data System (IMDS) management-level access (to include background rights). **(T-3).**

2.6.5.3. Brief scheduled actions and situations that could adversely impact the mission to chain of command. **(T-3).**

2.6.5.3.1. **(Added-AETC)** Ensure leadership is aware of all issues that may impact flight safety, equipment sustainment, or professional development. This will include information provided to the Airfield Operations Board (such as discrepancies/fix actions for critical system outages and>30 days nonprogression in training, shortfalls in task coverage, etc.). **(T-3)**

2.6.6. Utilize maintenance data and analysis products to solve problems and improve the maintenance effort. **(T-3).**

2.6.7. Using local procedures, inform the chain of command of problems beyond the capability to solve at the work center level. **(T-3).**

2.6.8. Obtain a CAFDEx/RSEP account to initiate AFTO Form 227. **(T-3).** For further information on submitting AFTO Form 227, See TO 00-25-108. **Note:** Not applicable to ANG units.

2.6.9. Submit requests for depot maintenance assistance (when required) in accordance with TO 00-25-108 to the appropriate branch within their MAJCOM, FOA, or DRU. **(T-3). Note:** Not applicable to ANG and AFR units. Depot Purchased Equipment Maintenance (DPEM) program is managed at the National Guard Bureau and AFR.

2.6.10. Review those portions of Operations Plans (OPLANs), Programming Plans (PPLANs) and Program Action Directives (PADs) that task the work center. **(T-3).**

2.6.10.1. Provide the work center capabilities to execute required tasks. **(T-3).**

2.6.10.2. Identify training, equipment, supply support, vehicle support, additional manpower or other key elements to execute required tasks. **(T-3).**

2.6.10.3. Ensure personnel are assigned to the correct Unit Type Code (UTC) and meet minimum training requirements. **(T-3).**

2.6.11. If the work center does NOT conduct 24-hour operations, provide a standby telephone number to organizations in which the work center supports. **(T-3).**

2.6.12. Conduct and document annual review of IMDS Equipment Inventory Listing to ensure all work center supported assets/resources are accurately reflected/listed in accordance with AFI 21-103, *Equipment Inventory, Status and Utilization Reporting*. **(T-3).**

2.6.13. Dispatch personnel with the technical data and support items needed to troubleshoot, repair, and restore systems in an expedient manner. **(T-3).**

2.6.14. Verify personnel receive flight line drivers training as required prior to entering aerodrome. **(T-3).**

2.6.15. Review the D04, Daily Document Register; D11, Commanders' Resources Information System 2 reports; Budget Code; Open Document Listing; and Daily Project Fund Management Record (PFMR)/Organizational Cost Center Record (OCCR) update and reconciliation reports to monitor work center expenditures. **(T-3). Note:** Supervisors should be aware of what portion of the maintenance budget was programmed for their use and how much has been expended. Supervisors should ensure funds are being used for their intended program (PEC 35111/Weather, PEC 35114/ATCALS). Work centers maintaining both weather systems and ATCALS should have an OCCR account (e.g., budgeted shop use, XB3, training/TDY expenses, Personal Protective Equipment, bench stock) and PFMR account (e.g., centrally managed funds, XD2, repair cycle assets) for each program. Reports can be obtained through the local supply customer service section.

2.6.16. Verify all Maintenance Data Collection (MDC) is accurate and complete. **(T-3).**

2.6.16.1. Document, maintain, and report system and/or equipment status in accordance with AFI 21-103 using IMDS. **(T-3).**

2.6.16.2. Use AFTO Form 349, *Maintenance Data Collection Record* if IMDS is temporarily unavailable and maintain a copy until IMDS is available. **(T-3).** Upon IMDS restoral, enter all applicable data as necessary. **(T-3).**

2.6.16.3. Review the Maintenance Action Review Report (MARR) to validate JDD weekly. **(T-3).** Review the Open Incident Listing (OIL) to validate Equipment Status Reporting (ESR) monthly against MARRs. **(T-3).**

2.6.16.4. Coordinate status changes with all concerned agencies and include start time and estimated time of return to operation (ETRO) in the job narrative or applicable IMDS field. **(T-3).** Update ETROs when the original cannot be met. **(T-3).**

2.6.16.5. Direct, document, and control removal/replacement/cannibalization/controlled substitution actions in accordance with TO 00-20-2. **(T-3).** For IMDS entries, use Air Force Computer Systems Manual (AFCSM) 21-561, Volume 2, *Maintenance Events*

Software User Manual (available at the following web site: **https://ceds.gunter.af.mil/Publications.aspx?AIS=35**) for documentation details.

2.6.16.6. Assign a Job Control Number (JCN) when a loss of capability or function in either equipment/system or service occurs. **(T-3).**

2.6.16.6.1. Issue JCNs to appropriate agency in accordance with AFI 21-103. **(T-3).**

2.6.16.6.2. Advise respective MAJCOM, FOA, or DRU of Not Mission Capable (NMC) outages lasting longer than 1-hour in accordance with MAJCOM, FOA, or DRU directives. **(T-2). Note:** This report is only for systems the MAJCOM, FOA, or DRU determined critical but do not require a COR.

2.6.16.6.3. Ensure the same JCN will be used on the job and/or applicable JDD/ESR throughout the life span of the job. **(T-3).**

2.6.16.6.4. Ensure JCN will remain open until the job is completed, regardless of whether or not control changes between the work center and other agencies. **(T-3).**

2.6.16.6.5. For outside agency support, capture customer service request tracking number in the IMDS job subject line/narrative (e.g., Civil Engineer Squadron (CES) work order number, AT&T tracking number, Fed Ex tracking number). **(T-3).**

2.6.16.6.6. Maintain the status of all active scheduled, unscheduled, and deferred jobs in IMDS. **(T-3).**

2.6.16.6.7. Document all work performed on equipment or actions in support of equipment (e.g. PMIs, Green/Amber/Red Jobs, TCTOs, Not Repairable This Station (NRTS) actions, etc.) accurately in IMDS. **(T-3).**

2.6.17. Track/update deferred PMIs to ensure timely completion. **(T-3).**

2.6.17.1. Update deferred PMIs via IMDS, or, if unavailable, the AFTO Form 349, in accordance with TO 00-20-2. **(T-3).**

2.6.17.2. PMIs that are deferred until the next PMI is due will not be cancelled. When the next PMI interval is scheduled, the new PMI may be cancelled with approval from the Flight Chief (e.g. 28-day PMI will be cancelled 27 calendar days after the original scheduled due date). **(T-3).** Do not cancel the original PMI. To cancel a subsequent PMI, document 1 minute of time in IMDS with the reason the PMI was not completed.

2.6.17.3. Obtain commander exercising operational control approval prior to cancelling PMIs twice consecutively. **(T-3).**

2.6.17.3.1. The PMI must be completed the second time. **(T-3**). If the PMI is not completed the second time, the maintenance activity will change the equipment status to NMC and notify the MAJCOM Functional Manager (MFM). **(T-3).**

2.6.17.3.2. If continued use of equipment is deemed necessary due to operational need, the commander exercising operational control assumes risks and full responsibility/accountability for damage to personnel/property during this directed period of equipment usage.

2.6.17.3.3. If waived by the commander exercising operational control, equipment status will remain NMC in IMDS. **(T-3).**

2.6.17.3.4. Waiver approval must be documented in IMDS. **(T-3).**

2.6.18. Submit a COR utilizing the template prescribed in **Table A7.1** to the corresponding MAJCOM, FOA, or DRU office and AFFSA System Management (**HQAFFSA.XMS@af.mil**) prior to shift change/end of the duty day for all systems listed in **Table A7.2** of this AFMAN. **(T-3).** The MFM may require a COR for additional systems. These requirements will be outlined in MAJCOM, FOA, and DRU directives. **(T-2). Note:** Not applicable to ANG DATCALs, Ground Tactical Air Controller (GTAC), and solar.

2.6.18. **(AETC)** Send CORs to the AETC Airfield Operations Workflow: **aetc.airfield.operationsworkflow@us.af.mil**.

2.6.19. Review and accomplish all TCTOs and document in IMDS. **(T-3).** Additionally, accomplish any other completion reporting instructions specified within the TCTO. **(T-3).** For IMDS TCTO processing and procedures, review AFCSM 21-568 Volume 2, *Time Compliance Technical Order (TCTO) Software User Manual*, found at **https://ceds.gunter.af.mil/Publications.aspx?AIS=35**. **Note:** Commanders are not authorized to waive the accomplishment of TCTOs but may waive the requirement of the work center supervisor from reviewing and accomplishing TCTOs if the Commander designates another office to perform that function. All TCTOs need to be documented in IMDS, regardless if they are applicable at location or not. TCTOs for RMC-supported equipment will be accomplished by or in coordination with RMC personnel. **(T-3).**

2.6.20. Utilize Restoral Priorities. **(T-3).** The work center will work jobs in such an order as to follow agreed-upon restoral priorities (e.g., work on a lower priority asset would be deferred if an outage on a higher priority asset occurs). **(T-3).** Guidance for establishing restoral priorities and outage response times is found in AFMAN 13-204, Volume 1, *Management of Airfield.* Generally, restoral priorities for key systems are determined with all key base/wing agencies. Many organizations also tie this to the base recovery plan. **Note:** DATCALS will establish priorities if necessary. **(T-3).**

2.6.21. Act as POC for depot and maintenance assistance requests in accordance with TO 00-25-108. **(T-3).**

2.6.22. Conduct and document annual review to ensure the master PMI Listing is up to date and to ensure all scheduled inspection requirements are being met per AFTO, FAA Technical Manual (TM), Commercial Manual, etc. **(T-3).**

2.6.23. Complete changes/updates to the master PMI Schedule within IMDS as required. **(T-3).**

2.6.24. Annually review all Service Level Agreements, Memorandums of Agreement, and/or Memorandums of Understanding with outside organizations as applicable. **(T-3).**

2.6.25. Establish recurring equipment downtime as necessary to facilitate OJT and PMI completion. **(T-3).**

2.6.26. Oversee the application of MICT checklists per MAJCOM, FOA, DRU, or wing guidance. **(T-3)**

2.6.27. Read and document completion of required reading documents for all personnel outlined in **Chapter 3** of this AFMAN annually and upon initial assignment or reassignment to a work center. **(T-3)**

2.6.28. Fulfill the role of support functions for the regionally maintained equipment as outlined in **Chapter 3** of this AFMAN. **(T-3)**

2.6.29. Maintain a work center safety program as outlined in AFMAN 91-203, AFI 91-202, *The US Air Force Mishap Prevention Program* and **Chapter 8** of this AFMAN. **(T-3).**

2.6.30. Work centers are required to perform Test, Measurement and Diagnostic Equipment (TMDE) user responsibilities in accordance with TO 00-20-14, *Air Force Metrology and Calibration Program*. **(T-3)**

2.7. Regional Maintenance Center

2.7.1. **Regional Maintenance Work Center (RMWC) will:**

2.7.1.1. Perform all routine and specialized maintenance for fixed base regionally maintained equipment to include resets, alignments, adjustments, and flight check inspections when required. **(T-3).**

2.7.1.2. Respond (by dialing into identified system) to troubleshoot maintenance alerts and system outages within 30 minutes during the normal duty hours and 60 minutes for non-duty hours, 24 hours per day, 7 days per week. **(T-3).**

2.7.1.3. Respond to system outages as first in, first out. **(T-3).** System restoral will be based on outage criticality, e.g., emergency outage vs. routine outage. **(T-3).**

2.7.1.4. For regionally maintained system outages, receive initial fault notification directly from the responsible ATC organization. **(T-3).**

2.7.1.5. Request permission to take control of the system and perform appropriate actions to correct the problem. **(T-3).** JCNs will be issued in accordance with AFI 21-103 to appropriate ATC agencies, as required. **(T-3).**

2.7.1.6. Coordinate with the local work center to order parts for regionally maintained equipment and notify the local maintenance unit of the expected delivery date. **(T-3).** Local maintenance personnel may conduct the parts replacement under the direction of RMWC personnel in accordance with TO 00-20-10, *Methods and Procedures-Air Traffic Control and Landing Systems Regional Equipment Maintenance Inspection, Documentation, Policies, and Procedures.*

2.7.1.7. Respond on-site as soon possible, when troubleshooting can no longer be completed remotely, in conjunction with the arrival of ordered parts, or if additional maintenance actions are required. **(T-3).**

2.7.1.8. Contact the appropriate ATC organization following restoral and return system control to the Watch Supervisor. **(T-3).** Additionally, a separate notification will be made to the RAWS work center if they were utilized to conduct maintenance actions. **(T-3).**

2.7.1.9. Visit each base/site annually to perform on-site maintenance/system optimization on regionally maintained systems. **(T-3).** RMWC's will coordinate visits with the work center. **(T-3).**

2.7.1.10. Perform acceptance evaluations during equipment installation of regionally maintained systems and provide orientation training to assigned work center personnel at that time. **(T-3).**

2.7.1.11. Notify Flight Inspection Coordination Office of out-of-cycle flight inspection needs due to maintenance actions. **(T-3).**

2.7.1.12. Provide outage updates via CORs to applicable customers, RAWS work center, and MAJCOM, FOA, or DRU within 1 hour (duty hours) or 2 hours (outside of normal duty hours). **(T-3).** Once a status change occurs, additional updates will be provided within the same timeframe. **(T-3).**

2.7.1.13. Provide a completed trip report for each site visited to MAJCOM/FOA/DRU and RAWS work center within 30 calendar days of return to home station. **(T-3).**

2.7.1.14. Hire personnel that meet 1C8X3 guidelines outlined in section 16 of the Special Duty Catalog (SPECAT). **(T-3).**

2.7.1.15. Utilize funds, facilities, and other resources in an effective and efficient manner and in the best interest of the RMWC. **(T-3).** Plan resource utilization, replenishment, and budget allocation to ensure personnel are provided the equipment and resources needed to accomplish the mission effectively. **(T-3).**

2.7.1.16. Provide RMWC technicians with applicable training (e.g., OJT, classroom) and maintain 100% work center task coverage. **(T-3).**

2.7.1.17. Develop/maintain duty, standby, and flight inspection support schedules to provide 24/7 operations. **(T-3).**

2.7.1.18. Verify all JDD/ESR is accurate and complete. **(T-3).**

2.7.1.19. Document, maintain, and report system and/or equipment status in accordance with AFI 21-103 using IMDS. **(T-3).**

2.7.1.19.1. Use AFTO Form 349 if IMDS is temporarily unavailable and maintain a copy until IMDS is available. **(T-3).** Upon IMDS restoral, enter all applicable data as necessary. **(T-3).**

2.7.1.19.2. Review the MARR to validate JDD weekly. **(T-3).** Review the OIL to validate ESR monthly against MARRs. **(T-3).**

2.7.1.19.3. Coordinate status changes with all concerned agencies and include start time and ETRO in the job narrative or applicable IMDS field. **(T-3).** Update ETROs when the original cannot be met. **(T-3).**

2.7.1.19.4. Direct, document, and control removal/replacement/cannibalization/controlled substitution actions in accordance with TO 00-20-2. **(T-3).** For IMDS entries, use AFCSM 21-561, Volume 2, (available at the following web site: **https://ceds.gunter.af.mil/Publications.aspx?AIS=35**) for documentation details .

2.7.1.19.5. Assign a JCN when a loss of capability or function in either equipment/system or service occurs. **(T-3).** The same JCN will be used on the job and/or applicable equipment status report throughout the life span of the job. **(T-3).**

2.7.1.19.5.1. The JCN will remain open until the job is completed, regardless of whether or not control changes between the work center and other agencies. **(T-3).**

2.7.1.19.5.2. For outside agency support, capture customer service request tracking

number in the IMDS job subject line/narrative (e.g., CES work order number, AT&T tracking number, Fed Ex tracking number). **(T-3).**

2.7.1.19.5.3. Maintain the status of all active scheduled, unscheduled, and deferred jobs in IMDS. **(T-3).**

2.7.1.20. Defer annual PMIs, when required, through the respective RMWC Superintendent. **(T-3).**

2.7.1.21. Support in the management, scheduling, and coordination of remote maintenance capable ATCALS installations. **(T-3).**

2.7.1.21.1. Develop comprehensive installation schedule in coordination with applicable Program Office and contractors. **(T-3).**

2.7.1.21.2. Assign qualified technicians to pertinent installations. **(T-3).**

2.7.1.21.3. Provide updates, as applicable, to the Program Office. **(T-3).**

2.7.1.21.4. Update customers with accurate/timely information on overall progress of the project. **(T-3).**

2.7.1.21.5. Verify required equipment data/documentation is gathered, accurate, and complete. **(T-3).**

2.7.1.22. Conduct site visits. **(T-3).**

2.7.1.22.1. Review and interpret all facility reference and site data prior to any site visit and equipment maintenance. **(T-3).**

2.7.1.22.2. Notify and coordinate all equipment downtime and Notice to Airmen (NOTAM) schedules with the RAWS work center supervisor. **(T-3).**

2.7.1.22.3. Notify and coordinate with local work centers for all assistance with remote or on-site maintenance on all affected systems. **(T-3).**

2.7.1.22.4. Conduct in-/out-briefs with respective operations agencies for all site visits. **(T-3).**

2.7.1.22.5. Collect and document all data on appropriate equipment forms after any maintenance is performed. **(T-3).**

2.7.1.22.6. Verify, document, and input all changes to facility reference data into IMDS, facility record, and any approved databases. (Verify by ground checks or necessary flight check.). **(T-3).**

2.7.1.22.7. Confirm equipment software configuration files are up to date and saved to any pre-designated file sharing locations. **(T-3).**

2.7.1.22.8. Perform preventive maintenance on regionally maintained equipment and initiate action to correct unsatisfactory equipment performance trends. **(T-3).** Record and maintain inspection and maintenance records. **(T-3).**

2.7.1.22.9. Provide applicable AFJQS training to a minimum of 1 trainer in the work center in conjunction with annual site visits **(T-3). Note:** The trainer will be responsible for training all other work center personnel in accordance with **Chapter 3** of this AFMAN. **(T-3).**

2.7.1.23. Assist in the development of AFJQS/AFQTPs for modernized airfield operations equipment and validate products within AETC to ensure standardized training and tasks lists for all maintenance elements supporting ATCALS. **(T-3).**

2.7.2. **Air Force Engineering and Technical Services Work Center will:**

2.7.2.1. Provide technical assistance, training, and emergency on-site assistance. Response to an on-site assistance request from the unit will be as soon as possible after request received.

2.7.2.2. Handle restoral priorities for non-remote systems on a first-in, first-out basis.

2.7.2.3. Notify the corresponding MAJCOM, FOA, or DRU when an AFETS technician responds to an on-site emergency assistance request.

2.7.2.4. Respond to and prioritize requests (based on mission impact, budget, and manning constraints) for on-site support. Owning units will request support via their respective MAJCOM, FOA, or DRU.

2.7.2.5. Provide telephone support to field technicians during normal duty hours.

2.7.2.6. Provide on-site outage response when outages exceed field technicians' abilities.

2.7.2.7. Provide training support on-site as requested.

2.7.2.8. Provide SME for analysis of complex problems, flight inspection failures, and engineering requests.

2.7.3. **Air Traffic Control and Landing Systems Radar Evaluation Work Center will:**

2.7.3.1. Provide first-level support to system outages in conjunction with radar AFETS, technical assistance, training, and emergency on-site assistance for terminal ATC radar systems. Initial response to an on-site assistance request will be as soon as possible after request is received from the MAJCOM. Coordinate restoral efforts if second-level assistance is required (e.g., FAA second-level engineering, Standard Terminal Automation Replacement System (STARS) Operational Support Facility). AFFSA is the approval authority for all requests for on-site second-level assistance.

2.7.3.2. Assist units with commissioning activities for new/relocated radar facilities. Duties may include coordination of geodetic survey; performing system performance analysis/optimization for mission requirements; planning and executing formal flight inspection; and documentation of results.

2.7.3.3. Perform baseline/periodic evaluations to define the operational capabilities and limitations of a facility and provide recommendations for optimum performance consistent with its assigned mission and operating environment. Based on historical data, these evaluations should be repeated every seven years or if there is a significant change in siting conditions, equipment configurations, assigned airspace, etc. Out-of-cycle requests will be forwarded through MAJCOM, FOA, or DRU.

2.7.3.4. Provide technical expertise to Program Office or ATCALS System Managers (SM) for technical instruction (TI) changes, system modifications, procedure development, and operational testing.

2.7.3.5. Re-optimize radar systems following changes to the system, radar environment, or mission (e.g., new construction, changes to assigned airspace).

2.7.3.6. Provide pre-construction analysis of potential radar impacts related to construction of new buildings, solar farms, and wind turbines in accordance with AFI 13-201, *Airspace Management*. Assist units with post-construction mitigation to reduce or eliminate impacts.

2.7.3.7. Monitor the FAA Obstruction Evaluation/Airport Airspace Analysis (OE/AAA) database for wind turbine projects in the vicinity of Continental United States (CONUS) terminal radars. Provide a representative to Air Staff Encroachment Management Working Group/Mitigation Response Team (EMWG/MRT) in accordance with AFI 90-2001, *Mission Sustainment*.

2.8. System Managers will:

2.8.1. Adhere to specific guidance for SMs that are assigned to a Lead Command as outlined in AFI 63-101/20-101, *Integrated Life Cycle Management.* **(T-3).**

2.8.2. Adhere to specific guidance for SM outlined in AFPD 10-9, *Lead Command Designation and Responsibilities for Weapon Systems.* **(T-3).**

2.8.3. Fulfill the responsibilities for SMs as outlined in **Chapter 6** of this AFMAN. **(T-3).**

Chapter 3

MAINTENANCE WORK CENTER

3.1. Introduction.

3.1.1. This chapter addresses maintenance work center guidelines, planning and scheduling, ordering and managing materials, and ensuring maintenance actions are properly documented and tracked from start to finish.

3.1.2. IMDS is the approved Automated Information System (AIS) for all maintenance activities. The fundamental concepts, duties, and procedures of a maintenance work center are detailed in the following paragraphs.

3.2. Self-Assessment Program.

3.2.1. Commanders will appoint a work center self-assessment program manager, who will oversee the work center's self-inspection/assessment program as outlined in AFI 90-201. **(T-3).**

3.2.2. Work center supervisors will oversee the application of MICT checklists per MAJCOM/FOA/DRU/wing guidance. **(T-3).** They shall provide all necessary documentation to the unit MICT manager, adhering to the Commander's Inspection Program. **(T-3).** Use **Attachment 3**, *Sampling Plan*, for all equipment inspections. **(T-3).**

3.3. Mandatory Annual Reading List.

3.3.1. Annually and upon initial assignment or reassignment, all RAWS technicians will read and document completion of:

3.3.1.1. This AFMAN. **(T-3).**

3.3.1.2. Applicable MAJCOM, FOA, or DRU supplements to this AFMAN. **(T-3).**

3.3.1.3. The 1C8X3 CFETP. **(T-3).**

3.3.2. Annually and upon initial assignment or reassignment, all RAWS technicians assigned to an ATCALS work center will read and document completion of the following:

3.3.2.1. AFMAN 11-225_IP, *United States Standard Flight Inspection Manual* (applicable sections for assigned equipment). **(T-3).**

3.3.2.2. TO 31Z3-822-2, *Air Traffic Control and Landing Systems (ATCALS) Site Requirements 404L*. **(T-3).**

3.3.2.3. AFMAN 13-204, Volume 3, *Air Traffic Control* (Applicable chapters). **(T-3).**

3.3.2.4. TO 00-20-10, if applicable. **(T-3).**

3.3.2.5. AFI 21-103. **(T-3).**

3.3.3. Annually it is recommended that RAWS technicians, based on unit of assignment, read:

3.3.3.1. AFI 13-217, *Drop Zone (DZ) and Landing Zone (LZ) Operations* (applicable chapters).

3.3.3.2. AFTTP 3-4.4, *Contingency Airfield Operations* (applicable chapters).

3.3.4. Upon initial assignment, all Non-Commissioned Officer in Charge will read AFQTP 1C8X3-201D and annually thereafter. **(T-3).** This will also be required for any Airman assuming the roles of NCOIC for greater than 3 months, regardless of duty title. **(T-3).** All other Airmen will read AFQTP 1C8X3-201D as part of their 7-level upgrade. **(T-3).**

3.4. Air Traffic Control and Landing Systems Facility Requirements.

3.4.1. Equipment TOs/TMs may require PMI data record entries by qualified technicians on a recurring basis.

3.4.2. Documentation of the completion of these PMIs or other required performance measurements will meet the ATCALS facility references. **(T-3).**

3.4.3. Fully qualified technicians performing PMI procedures will meet all requirements to validate key parameters against the facility references, TO/TM specifications, or AFMSIs. **(T-3).**

3.4.4. ATCALS consists of ATC radar, navigational, and air traffic radio, console, and recorder technologies.

3.5. Air Traffic Control and Landing Systems Facility Records.

3.5.1. Maintenance work centers will maintain a facility record at each ATCALS facility (may not be applicable to radio sites). **(T-3).** This record is a transitory portion of the equipment historical file. The site applicable forms prescribed in TO 31Z3-822-2 and system specific certification forms, when outlined by equipment TOs/TMs, are mandatory and must be completely filled out. **(T-3).** Superseded data will be retained in the work center equipment historical file. **(T-3).** The RMC will maintain facility records for equipment maintained by the RMC. **(T-3).** These records will be kept at the RMC, not the local site. **(T-3).** Electronic facility records are authorized.

3.5.2. Facility record will contain the following information:

3.5.2.1. Tab 1, *Reference Data*. Contains the reference flight inspection report and all data recorded following the reference flight inspection. **(T-3).** Each page must be clearly marked at the top, "REFERENCE DATA," with the effective date of the reference data. **(T-3).** When references change as a result of maintenance, parts replacements, etc., make a single line in red pen through the changed item on the original reference data worksheet, annotate the new reference, and add the date the change was made. **(T-3).** When the reference values are changed, explain the reason for the change in any of the available remarks' blocks. **(T-3).**

3.5.2.2. Tab 2, *Preventive Maintenance Inspection*. Contains data recorded during PMIs as required by applicable maintenance TO/TMs. All required blocks on the appropriate forms must be completed. **(T-3).** An explanation is required in the remarks section of the form if required data is not recorded. **(T-3).** The remarks section of the form may be used to document trends identified/adjustments made during PMIs. If adjustments are made during the PMI, the final reading will be recorded. **(T-3).** The most recent piece of certification data will be kept in the facility record and the previous two years of certification data will be maintained in equipment historical records. **(T-3).** For command unique equipment, use local, MAJCOM FOA, or DRU generated forms. **(T-3).**

3.5.2.3. Tab 3, *Periodic Flight Inspection Reports*. Contains the last two periodic flight inspection reports and the FAA Form 8240-22, *Facility Data,* or electronic 7900-series equivalent (validate accuracy and currency with terminal instrument procedures). **(T-3).**

3.5.2.4. Tab 4, *PMI Ground Check*. Contains data recorded during PMIs to determine radiation characteristics as required by TO/TMs. **(T-3).**

3.5.2.5. Tab 5, *Other Data*. Include pertinent site information not filed elsewhere. **(T-3).** Most recent CES grounding report is required. **(T-3).** Optional items include the most recent: mobile depot maintenance, completed AFMQCCs, RMC trip report, ATCALS Evaluation, AFETS trip report and earth ground check documentation.

3.6. Air Traffic Control and Landing Systems Facility Reference Data.

3.6.1. If the facility cannot be returned to the established facility references, but facility performance meets or exceeds TO specifications and can be verified with a ground check, the facility is considered usable.

3.6.2. Work centers will perform complete data collection and forward existing facility reference data and newly recorded data to the RMC within 5 duty days for analysis. **(T-3).**

3.6.3. The RMC will determine if further adjustments are required, or if a special (reference) flight inspection must be accomplished to establish new facility references. **(T-3).**

3.6.4. If proper facility performance cannot be verified with a ground check, work centers will report the facility's condition to the senior ATC supervisor as unusable and request RMC assistance through the MAJCOM, FOA, or DRU. **(T-3).**

3.6.5. All systems subject to FAA flight inspection (e.g. Tactical Air Navigation System, Instrument Landing System, Digital Airport Surveillance Radar) will meet inspection requirements prior to being placed into service in accordance with AFMAN 11-225_IP/FAA Order 8200.1D. **(T-0). Note:** AFMAN 11-225, authorizes military installation commanders or SAA at a Joint Base to publish NOTAMs for facilities not part of the NAS as "For Military Use Only" when a facility has not received commissioning or subsequent inspections due date window(s).

3.7. Air Traffic Control and Landing Systems Maintenance Special Considerations for Removing Navigational Aid Identification Signals.

3.7.1. Aircrews are trained to consider the navigation signal unreliable when Identification (ID) is not received or the ID code is "T E S T" (-. … -).

3.7.2. The removal of ID will not be a substitute for NOTAM requirements. **(T-1).** Technician must remove ID or transmit "T E S T" ID when the Navigational Aid (NAVAID) facility is out of service. **(T-1).**

3.7.3. The preferred method is to remove ID when performing maintenance.

3.8. Other Equipment.

3.8.1. Work centers maintaining equipment that they do not have ownership of will obtain a local MOA or SOW for maintenance. **(T-3).**

3.8.2. Review and document MOA or SOW annually.

3.9. Work Center functions in support of Regionally Maintained Equipment. Work centers supporting RMCs must:

3.9.1. Provide a 24-hour POC to the RMC. **(T-3).**

3.9.2. Work directly with the RMC to replace failed assemblies in accordance with TO 00-20-10. **(T-3).**

3.9.3. Respond to outages, after being notified by the RMC, on a 24-hour basis. **(T-3).**

3.9.4. Identify a point of receipt for parts delivery. **(T-3).**

3.9.5. Coordinate with RMC for return of failed assemblies. **(T-3).**

3.9.6. Work directly with the RMC to perform tasks in accordance with TO 00-20-10, as well as general maintenance practices. **(T-3).**

3.9.7. Serve as RMC liaison for on-site telephone, status line or modem line problems using local procedures. **(T-3).** Furthermore, coordinate with host base support agencies (e.g. CS, CES, Logistics Readiness Squadron (LRS) to resolve any issues necessary to facilitate remote monitoring or maintenance for the RMC. **(T-3).**

3.9.8. Work directly with CES to resolve commercial power or Environmental Control Unit problems (including grounding and lightning protection). **(T-3).** If modification is required, coordinate with the appropriate MAJCOM, FOA, or DRU branch in accordance with **Chapter 10** of this AFMAN.

3.9.9. Provide facility access, flight line communication, qualified flight line driver, local technicians, and local hazard briefing to assist with system maintenance to the RMC technicians when on-site maintenance is performed. **(T-3).**

3.9.10. Perform escort duties and provide facility access for required local support agencies (e.g. local CES, Safety). **(T-3).**

3.9.11. Maintain common hand tools, test equipment, and climbing gear to assist RMC technicians in accordance with TO 00-20-10. **(T-3).**

3.9.12. Maintain accountability for the regionally maintained equipment on Custodian Authorization Custody Receipt Listing (CA/CRL). **(T-3).**

3.9.13. Verify TMDE is properly calibrated and any limited calibrations meet the system requirements. **(T-3).** For assistance with limited calibration contact the RMC. **(T-3).**

3.9.14. Perform facility/building manager duties to include DPEM and CAFDEx/RSEP requests on shelters, antenna towers, etc., in accordance with TO 00-25-108. **(T-3).**

3.9.15. Perform general housekeeping and corrosion prevention and control duties in accordance with applicable equipment TO's on regionally maintained systems that do not require powering down of system. **(T-3).** If required to power down system, coordinate actions with servicing RMC. **(T-3).**

3.9.16. Coordinate downtime requests on behalf of the RMC in accordance with local procedures. **(T-3).** Provide approval confirmation via e-mail to the RMC organizational e-mail account. **(T-3).**

3.9.17. Create/maintain an organizational e-mail account to coordinate official communication between the RMC and local work center. **(T-3).**

3.9.18. Load applicable AFJQSs for modernized systems to the MTP. **(T-3).**

3.9.19. Perform only tasks that work center personnel are qualified to perform and only do so when directed by RMC personnel. **(T-3).**

3.10. ATC-Radio Antenna PMIs.

3.10.1. Qualified RAWS technicians will conduct PMIs on all ATC-radio antennas. **(T-3).**

3.10.2. Units without the capability or technical experience to perform PMIs will submit an AFTO Form 229**,** *Engineering Installation Assistance Request,* or request assistance from the local cable and antenna team. **(T-3).**

3.10.3. PMIs will be completed in accordance with governing publications to include: AFMSI 300-1, *Scheduled Periodic Inspections (336 Day Interval) For Air Traffic Control Communications System,* TO 31R-10-5, *Air Force Communication Commands (E-I Standard) - Antenna Systems, Maintenance, Repair and Testing*, TO 31-10-21, *Installation Practices: Air Force Communications Service (E-I Standard) Antenna System Protection, Stepping, and Splicing of Poles*, and applicable commercial manuals. **(T-3).**

Chapter 4

MOBILITY WORK CENTER

4.1. Introduction.

4.1.1. This chapter addresses maintenance management policies for all units with a mobility mission. Units, regardless of their organizational structure, shall follow the principles of maintenance management to ensure efficient and effective maintenance. **(T-3).**

4.1.2. All units with a deployable RAWS mission and deployable UTC, will apply and perform the management policies prescribed within this AFMAN unless otherwise specified. **(T-3).**

4.1.3. UTC based work centers must abide by this chapter, as well as the other applicable chapters of this AFMAN. **(T-3).**

4.1.4. In-garrison maintenance, pre-deployment planning, and pre-deployment preparation are critical to ensuring equipment and personnel readiness when tasked for contingency operations. Maintenance management procedures in a deployed environment must be appropriately scaled to balance maintenance requirements against operational commitments. **(T-3).** Personnel at all levels must leverage opportunities to compensate for the inefficiencies of contingency operations. **(T-3).**

4.1.5. Units with primarily deployable missions will designate a 1C8 Team Lead prior to deployment and support management functions and activities. **(T-3).** The appointed 1C8 Team Lead, managing functions structured by the UTC, will ensure these functions adhere to the principles of a maintenance work center in accordance with **Chapters 3** of this AFMAN. **(T-3).**

4.1.6. The deployed Chief of Maintenance (COM) or Superintendent is the authority for overall management, while equipment/UTC Team Leads will ensure their personnel comply with established guidance. **(T-3).** Deployed maintenance management focuses on delivering the required operational capabilities while limiting operation interruptions.

4.2. Chief of Maintenance. The COM will plan, organize, staff, direct, and manage the maintenance effort and is responsible to the commander for accomplishing the maintenance mission. **(T-3).** The COM will use the following procedures to achieve the highest level of maintenance effectiveness. **(T-3).** As a minimum, the COM will:

4.2.1. Adhere to duties and responsibilities of a flight commander as outlined in **Chapter 2** of this AFMAN. **(T-3).**

4.2.2. Manage civil service employees according to Office of Personnel Management and Air Force directives. **(T-3).**

4.2.3. Temporarily realign duties to ensure efficient use of assigned manpower. **(T-3).**

4.2.4. Create and implement a local Maintenance Operation Instruction. **(T-3).**

4.2.5. Review and evaluate maintenance performance and training effectiveness with the commander, maintenance superintendent(s), team chiefs, QA and maintenance supervisors

quarterly using the deficiency analysis summary, maintenance training summary, and/or other relevant management products. **(T-3).**

4.2.6. Review Personnel Evaluation (PE) reports that document unsatisfactory task results. **(T-3).**

4.2.7. Approve QA-validated LWC for publication and use by signing the LWC title page. **(T-3).**

4.2.8. Establish effective training programs in all maintenance work centers, to include but not limited to RAWS, QA/Maintenance Operations Center (MOC), material control/supply, and squadron Civil Engineering (CE). **(T-3).**

4.2.9. Develop and implement life cycle logistics support plans for unit acquired COTS systems and equipment. **(T-3).**

4.2.10. Establish effective safety and radiation protection practices according to AFMAN 91-203, and TO 31Z-10-4, *Electromagnetic Radiation Hazards*, Appendix A. **(T-3).**

4.2.11. Establish close working relationships with base support activities. **(T-3).**

4.2.12. Provide accurate visibility and documentation of equipment assets in IMDS. **(T-3).**

4.2.13. Provide accurate visibility and documentation of personnel upgrade, proficiency and cross-utilization training in the Training Business Area (TBA). **(T-3).**

4.2.14. Provide HAF/MAJCOM/FOA/DRU -level staff with timely and accurate equipment specific and/or personnel issues that can or will be a hindrance to the unit's mission. **(T-3).** Coordinate actions that impact airfield operations with the local SAA, Air Operations Flight Commander, Air Traffic Manager, or equivalent level of authority. **(T-3).**

4.3. Quality Assurance.

4.3.1. QA provides the oversight, evaluation and compliance of maintenance programs and equipment. QA will provide status of the program to the COM. **(T-3).** The QA work center's purpose is to identify any area, program of equipment/system that is not compliant with standards; identify the deficiency, root cause and possible solutions; analyze to determine if there are trends and provide a comprehensive report to all levels of management or identify any area, program or equipment/system that is exceeding standards, benchmark them and report to all levels of management. The COM actively manages the QA work center/programs. As a minimum, the QA work center will:

4.3.1.1. Provide unit leadership with an analysis, reporting, and deficiency resolution capability. **(T-3).**

4.3.1.2. Assist the COM in ensuring all maintenance processes, equipment, systems, end items or services are of the type and quality to meet or exceed requirements for effective mission operations. **(T-3).**

4.3.2. QA will be the lead for the following functions:

4.3.2.1. I&E program conducted in accordance with **Chapter 7** of this AFMAN. **(T-3).**

4.3.2.2. Self-Inspection/Assessment Program Manager. **(T-3).**

4.3.2.3. Corrosion Prevention and Control Program (CPCP) Monitor. **(T-3).**

4.3.2.4. Electrostatic Discharge (ESD) Monitor. **(T-3).**

4.3.2.5. Conducts work center ESD Control surveys according to TO 00-25-234, *General Shop Practice Requirements for the Repair, Maintenance and Test of Electrical Equipment*. **(T-3).**

4.3.2.6. Technical Order Distribution Office (TODO) in accordance with **Chapter 12** of this AFMAN. **(T-3).**

4.3.2.7. MICT administrator/manager. **(T-3). Note:** The QA NCOIC should be considered for a wing inspection team position for RAWS functions.

4.3.2.8. Unit Lockout/Tag Out (LOTO) Monitor in accordance with **Chapter 8** of this AFMAN. **(T-3).**

4.3.2.9. Evaluate the quality of maintenance performed in the maintenance work centers. **(T-3).**

4.3.2.10. Ensure completion of Deficiency Reports in accordance with TO 00-35D-54, *USAF Deficiency Reporting, Investigation, and Resolution*. **(T-3).**

4.3.2.11. Perform technical reviews of modification proposals and process valid proposals according to applicable directives. **(T-3).**

4.3.2.12. Validate LWC, when required, in accordance with **Chapter 12** of this AFMAN. **(T-3).**

4.3.2.13. Train Quality Assurance Representatives (QARs) and spot check QAR performance. **(T-3).**

4.3.2.14. Maintain a record copy of all applicable, published AFMQCCs. **(T-3).** AFMQCCs can be used in the absence of an Air Force -wide standardized checklist or governing DD Form for a particular process, function, or piece of equipment.

4.3.2.15. AFMSIs can be accessed on the RAWS Operations, Training, and Procedures SharePoint®. See **Attachment 5** of this AFMAN for a listing of AFMSIs and link to SharePoint®.

4.4. Maintenance Operations Center.

4.4.1. Units will establish a 24-hour POC for each deployed equipment/UTC and ensure they perform MOC functions. **(T-3).**

4.4.2. All other duties and responsibilities will be outlined in a unit Operating Instruction (OI). **(T-3).**

4.4.3. Units will outline in-garrison MOC functions in the unit OI. **(T-3).**

4.5. Materiel Control. Materiel Control will function as organizational supply and provide mobility support. **(T-3).** Materiel Control will perform the following duties:

4.5.1. Set up a supply and equipment account with the host base as identified in the tasking directive. **(T-3).** This will include the ID of contact points, means of communication, and delivery destinations. **(T-3).**

4.5.2. Identify deployed or transferred assets to the support base. **(T-3).** For long-term deployments, transfer assets to the support base account via Redistribution Order procedures. **(T-3).**

4.5.3. Follow Readiness Spares Package (RSP) storage availability, issue and turn-in procedures, and establish procedures for RSP replenishment at deployed locations. **(T-3).**

4.5.4. Maintain a "part number to National Stock Number" cross-reference capability on all items contained in the RSP(s). **(T-3).** Web Federal Logistics Information System (WEBFLIS)/Federal Logistics (FEDLOG) is recommended.

4.5.5. Close supply and equipment accounts with the support base upon termination of deployment. **(T-3).**

4.6. 1C8 Unit Type Code Team Lead. The Team Lead will adhere to and enforce all maintenance work center functions. **(T-3).** In addition, Team Leads will perform the following duties:

4.6.1. Verify completion of pre-deployment and post-deployment inspections for assigned UTC to include:

4.6.1.1. Completion of highest interval preventive maintenance inspection and below. **(T-3).**

4.6.1.2. Completion of End Item equipment inventory AFTO Form 470, *Electronic Set Inventory Checklist*, AFTO Form 471, *Electronic Set Inventory Checklist Configuration Data*, and AFTO Form 472, *Electronic Set Inventory Checklist Completion Data*, and include copies of all historical records. **(T-3).**

4.6.2. Check UTC inventory against Logistics Detail for completion and accuracy. **(T-3).**

4.6.3. Check RSP inventory for completion and accuracy. **(T-3).**

4.6.4. Complete all required inspections on UTC assigned Primary Movers and Mobilizers. **(T-3).**

4.6.5. Verify all cargo is packed, prepared, and processed for shipment in accordance with AFI 10-403, *Deployment Planning and Execution*. **(T-3).**

4.6.6. Ensure all required shipping documents are acquired. **(T-3).**

4.6.7. Coordinate with appropriate work centers to ensure all support equipment will be available and in place (e.g. Power Pro, Heating, Ventilation, Air Conditioning, and Vehicles). **(T-3).**

4.6.8. Ensure all TMs are current, complete, and packed. **(T-3).**

4.6.9. Verify all required maintenance documentation forms and supply forms are acquired and packed. **(T-3).**

4.6.10. Verify all members of UTC follow established maintenance and safety practices in accordance with this AFMAN. **(T-3).**

4.6.11. Verify all maintenance actions are properly documented in IMDS or on an AFTO Form 349 until IMDS can be updated. **(T-3).**

4.6.12. Verify documentation of all critical maintenance actions in historical records. **(T-3).**

4.6.13. Verify equipment statuses are being reported in accordance with AFI 21-103. **(T-3).**

4.6.14. Verify completion of Maintenance Operation Center responsibilities if assigned by the site commander. **(T-3).**

4.7. Inspections and Evaluations.

4.7.1. In garrison, work centers, equipment, and personnel will be evaluated under the UTC evaluation requirements in **Chapter 7** of this AFMAN. **(T-3).**

4.7.2. When deployed, in most cases, UTC evaluations are not required. However, reconstitution and redeployment of equipment/UTCs may require accomplishment of evaluations. **Note:** If evaluations are performed, the following I&E duties will be accomplished at the discretion of the deployed unit commander:

4.7.2.1. Conduct or participate in site transfer inspections when relieving/replacing personnel at a deployed location. **(T-3).** This will only be done if previously deployed equipment/UTCs will remain in place at the deployed location. **(T-3).** Site transfer inspections ensure systems are operational, problems are identified, and equipment inventory is complete and capable of meeting mission requirements. The deployed site commander will determine the personnel best suited to perform the evaluations. **(T-3).** The evaluation report will be routed and closed within the deployed unit. **(T-3).**

4.7.2.2. Document site transfer inspections within 15 calendar days after assuming maintenance responsibility. **(T-3).** If possible, perform inspection with incoming and outgoing personnel. During inspections, the evaluator will accomplish the following as a minimum:

4.7.2.2.1. Accomplish minimum essential checks to ensure the systems are operating within required parameters in accordance with associated technical data. **(T-3).**

4.7.2.2.2. Verify all safety items are on-hand and safety deficiencies identified. **(T-3).**

4.7.2.2.3. Verify all Communication-Electronics (C-E) facilities, shelters, work benches, and systems are grounded properly. **(T-3).**

4.7.2.2.4. Conduct corrosion prevention and control as required. **(T-3).**

4.7.2.2.5. Verify all mobility markings are up-to-date. **(T-3).**

4.7.2.2.6. Verify required support items (e.g., technical data, tools, TMDE, RSP, are available and in proper condition to support a sustained deployment.) **(T-3).**

4.7.2.2.7. Verify master PMI Schedule and all required PMIs are scheduled and performed at appropriate interval. **(T-3).**

4.7.2.2.8. Account for all assets and verify documentation on applicable CA/CRL, mission critical systems and equipment listings, or other equipment accountability and inventory documents. **(T-3).**

4.7.2.2.9. Maintain a copy of the site transfer inspection on site for a minimum of 12 months. **(T-3).**

4.7.3. QA evaluators will perform air mobility and road mobility evaluation for redeploying equipment. **(T-3).** These evaluations must be completed prior to redeployment of equipment/UTCs to home station or alternate deployment location. **(T-3).**

Chapter 5

REGIONAL MAINTENANCE CENTER

5.1. Introduction . This chapter describes procedures and guidelines to ensure safe and reliable ATCALS performance for both regionally and non-regionally maintained systems. The RMC consists of the RMWC, the AFETS work center, the Radar Evaluations work center, and the Geodetic Survey work center. This chapter outlines RMC support of local work centers and systems in the European, Pacific, and CONUS theaters. AFFSA manages all three RMCs in support of the using commands and local units.

5.2. RMC Locations. The CONUS RMC is located in Oklahoma City, OK, and services the continental U.S. as well as Greenland, and South/Central America. The European RMC is located at Kapaun AS, Germany, and services sites in Europe and Africa. The Pacific RMC is located at Yokota AB, Japan, and services sites in the Pacific, including Alaska and Hawaii.

5.3. RMC Responsibilities. The respective RMC provides maintenance support for remote maintenance systems. All RMCs provide on-site maintenance for equipment within their respective region but may be called upon to support other regions. The RMCs provides traditional AFETS and ATCALS Radar Evaluation support for all units. See **Chapter 2** and **Chapter 3** for RMC functional responsibilities.

Chapter 6

SYSTEM MANAGERS

6.1. Introduction.

6.1.1. This chapter outlines specific functions for SMs that are assigned to a Lead Command. See AFI 63-101/20-101**,** for Lead Command definition. SMs provide an Air Force focal-point for maintenance and sustainment support for ground communications electronics equipment/systems.

6.1.2. SMs ensure common systems are reliable, maintainable, and available to meet mission needs by providing all levels of support.

6.1.3. SMs are in direct communication with the field units, Program Office, ATCALS System Program Office, logistics and acquisition communities, depots, Lead Command, Cyber Systems Integrators, AFETS, Engineering and Installation (E&I) community, the MAJCOM, FOA, DRU Representatives, field level SMEs, and AFCFMs.

6.2. System Manager Functions.

6.2.1. Basic system management functions will include, but are not limited to, those identified in AFPD 10-9. **(T-3).**

6.2.2. In addition, SMs will:

6.2.2.1. Work with the MAJCOM, FOA, or DRU to ensure equipment/systems are available to meet mission requirements. **(T-3).**

6.2.2.2. Understand the reason for, and role of, designated Lead Command and develop/maintain close working relationships with Lead Commands for their designated systems. **(T-3).**

6.2.2.3. Develop and maintain a POC list for all Item/Program Managers (PM), MAJCOM, FOA, or DRU Staff, and all maintenance work centers within each MAJCOM, FOA, or DRU. **(T-3).** Use of an organizational email address is the preferred contact method for initial e-mail correspondence to and from SM.

6.2.2.4. Stay abreast of related technological changes/modifications in order to effectively manage the equipment/systems for which they are responsible. **(T-3).**

6.2.2.5. Address Higher Headquarters (HHQ) inquiries and act as SME to support system development, sustainment, and logistical support. **(T-3).** Research, develop courses of action, and provide recommendations on life cycle management of fielded systems. **(T-3).**

6.2.2.6. Provide logistical SME support by conducting reviews of Air Force Equipment Management System (AFEMS), Allowance Standard Code, and other supply approved AIS. **(T-3).** Provide change/improvement recommendation to MAJCOM, FOA, DRUs or individual organizations. **(T-3).**

6.2.2.7. Interpret equipment-specific policy and procedures for requesting organizations. **(T-3).**

6.2.2.8. Assist with development of AFMQCC, AFMSIs, other inspection checklists, and policy. **(T-3).** See **Attachment 4** and **Attachment 5** for a list of AFMQCCs and AFMSIs.

6.2.2.9. Participate in maintenance and logistics community working groups, conferences, and seminars. **(T-3).**

6.2.2.10. Conduct data calls for Lead Commands and on behalf of HHQ or a MFM. **(T-3).** Draft requests, validate responses, consolidate information, conduct data/root cause analysis, and provide detailed report. **(T-3).**

6.2.2.11. Assist in revision of system policy, training, and guidance documents (e.g., TO, AFJQS, AFQTP, AFI, CFETP, as well as other documents.) **(T-3).**

6.2.2.12. Review system/equipment documentation for policy and procedural implications and coordinate legal review during acquisitions process. **(T-3).**

6.2.2.13. Coordinate and resolve MDC issues with applicable organizations. **(T-3).**

6.2.2.14. Coordinate operational issues that affect mission degradation with lateral agencies (e.g., Lead Command, Program Office, Primary Contracting Officer, FOAs, 38 Cyberspace Engineering Group, AFETS, and associated system depot offices). **(T-3).**

6.2.2.15. Assist with new system requirements from Lead Command, MAJCOM, FOA, DRU or unit. SM shall coordinate requirements with appropriate agencies, provide a comprehensive recommendation, and forward validated requirement to appropriate office. **(T-3).**

6.2.2.16. Coordinate and process system modification request per TO 00-5-1 and AFI 63-101/20-101. **(T-3).**

6.2.2.17. Provide SME support on system/equipment Life Cycle Management Plans (LCMP). **(T-3).**

6.2.2.18. Provide technical input, requirements, and assessment on system/equipment contracts. **(T-3).** SMs may also provide source selection recommendation/technical assessment.

6.2.2.19. Review system/equipment architecture and technical information from equipment manufacturer/lateral agency to ensure fielded systems are reliable, available to meet mission needs, and logistically supportable. **(T-3).**

6.2.2.20. Validate discrepancy notices and recommend potential solutions. **(T-3).**

6.3. System Manager Points of Contact.

6.3.1. AFFSA serves as the SM POC for Instrument Landing System (ILS), Tactical Air Navigation (TACAN), Very High Frequency Omnidirectional Range Tactical Air Navigation (VORTAC), ATCALS Radars, DATCALS and FAA-managed systems such as Enhanced Terminal Voice Switch (ETVS), Digital Audio Legal Recorder (DALR), Digital Air Surveillance Radar (DASR), STARS, etc.

6.3.2. Units requiring assistance for non-ATCALS/DATCALS systems, command-unique, or cross-utilized end items will contact the MFM. **(T-3).**

6.3.3. Units requiring Next Generation Radar (NEXRAD) assistance contact Air Force Materiel Command Instruction (AFMC)/AF557 WW/HBAW-OL. See **Attachment 6** of this AFMAN for contact information.

Chapter 7

INSPECTIONS AND EVALUATIONS

7.1. Introduction.

7.1.1. The term Inspections and Evaluations (I&E) refers to Standardization and Evaluations (Stan/Eval) and Quality Assurance (QA). Each unit will provide a minimum of one RAWS technician to perform the I&E function. **(T-3).** The RAWs Technician's primary duty is to ensure mission accomplishment occurs within the confines of law **(T-0)**, DoD policy and guidance **(T-0)**, Air Force policy and TOs. **(T-3).** I&E also provides an analysis, reporting, and deficiency resolution capability. It serves as the tool for ensuring that a process, equipment, system, end item, or service is of the type and quality to meet or exceed requirements for effective mission operations.

7.1.2. I&E is applicable to all field-level units with RAWS personnel. Those units will apply and perform the I&E policies prescribed within this AFMAN and applicable checklists. **(T-3).** Unit leadership is charged with the safety and quality of equipment operation, training, and maintenance. These actions will utilize the Air Force Inspection System (AFIS) self-assessment program as detailed in AFI 90-201, to meet compliance and evaluation requirements. **(T-3).**

7.2. Inspection & Evaluation Functions.

7.2.1. The unit commander will appoint a minimum of one RAWS technician to serve as the I&E evaluator. **(T-3).** Staff functions will be evaluated in accordance with the AFIS. **(T-3).**

7.2.2. All I&E evaluators will:

7.2.2.1. Be fully qualified on 100% of the tasks contained in the work center MTP. Assigned military personnel will have a minimum primary AFSC at the 7-skill level or civilian equivalent. **(T-3).**

7.2.2.2. Complete AFJQS 3DXXX-201G, *Quality Assurance*, within 3 months of assignment. **(T-3).**

7.2.2.3. Complete the 1C8X3-201D if not previously completed. **(T-3).**

7.2.2.4. Not be a trainer in the RAWS maintenance work center. **(T-3).**

7.2.2.5. Not be the RAWS self-inspection program manager. **(T-3).**

7.2.2.6. Not be assigned to the RAWS maintenance work center. **(T-3).**

7.2.2.6.1. I&E representatives may be assigned to the RAWS maintenance work center. **Note:** I&E representatives are members of the work center, not permanently assigned as an I&E evaluator. These I&E representatives will report directly to the flight commander for all I&E evaluations. **(T-3).**

7.2.2.6.2. I&E representatives must meet the training requirements listed in **paragraph 7.2.2** thru 7.2.5. **(T-3).**

7.2.2.7. Not be the RAWS self-inspection program manager. **(T-3).**

7.2.3. I&E evaluators are encouraged to complete in-residence QA course within 6 months of assignment.

7.2.4. I&E evaluators should serve as evaluators for a maximum of 24 months. Additionally, 24 months should pass before being allowed to serve again as I&E evaluator.

7.2.5. I&E evaluators may be civilian personnel.

7.2.6. I&E evaluators are encouraged to complete in-residence QA course within 6 months of assignment.

7.2.7. Fixed-Base Units I&E will perform the following functions:

7.2.7.1. Inspections and Evaluations in accordance with **Chapter 7** of this AFMAN. **(T-3).**

7.2.7.2. Corrosion Prevention and Control Program (CPCP) Monitor. **(T-3).**

7.2.7.3. Electrostatic Discharge (ESD) Monitor. **(T-3).**

7.2.7.4. Conducting work center ESD Control surveys according to TO 00-25-234. **(T-3).**

7.2.7.5. Technical Order Distribution Office (TODO) in accordance with **Chapter 12** of this AFMAN. **(T-3).**

7.2.7.6. Unit Lockout/Tag Out (LOTO) Monitor in accordance with **Chapter 8** of this AFMAN. **(T-3).**

7.2.7.7. Evaluate the quality of training and maintenance performed in the maintenance work centers through the use of Personnel and Equipment Evaluations (EE). **(T-3).**

7.2.7.8. Ensure completion of Deficiency Reports in accordance with AFI 23-101, *Air Force Material Management*. **(T-3).**

7.2.7.9. Perform technical reviews of modification proposals and process valid proposals according to applicable directives. **(T-3).**

7.2.7.10. Validate LWC, when required, in accordance with **Chapter 12** of this AFMAN. **(T-3).**

7.2.8. Mobile Units UTC evaluations will consist of three separate components designed to measure readiness: personnel readiness evaluation, equipment readiness evaluation, and acceptance readiness inspection when new or overhauled equipment is received from the manufacturer/depot. **(T-3).**

7.3. Managerial Evaluations. Managerial Evaluations will be performed as part of the AFIS. **(T-3).**

7.4. Personnel Evaluations.

7.4.1. The Personnel Evaluation (PE) program assesses the effectiveness of a work center's training program, technician competence, and technical and procedural data adequacy. These evaluations ensure equipment or systems are maintained in an effective and efficient manner to meet mission requirements. The evaluation can be accomplished on a system, equipment, or service-oriented function the technician is qualified to perform (e.g., align radar system, repair navigation system, follow a written process). Evaluators observe how well processes

are performed to determine if the technician demonstrates sufficient skill to accomplish the tasks.

7.4.2. There are four types of PEs: Individual Personnel Evaluations (IPE), UTC Evaluations, Special Personnel Evaluations (SPE), and Equipment Personnel Evaluations (EPE).

7.4.2.1. **Individual Personnel Evaluation.** 1C8 personnel shall complete within one year of initial assignment to an organization and upon return to the maintenance work center after an absence of one year or longer. **(T-3).** Short tour overseas assignments of 18 months or less, 1C8 personnel will accomplish the IPE no later than 9 months' time on station. **(T-3).** 1C8 personnel will accomplish recurring IPEs will be accomplished on a 12-month cycle. **(T-3).** This does not apply to MSgts - CMSgts unless they maintain training records and are task qualified to maintain equipment. AFR and ANG personnel shall conduct this evaluation within two years of assignment. **(T-3).**

7.4.2.2. **Unit Type Code Personal Evaluation.** UTC PEs assess UTC personnel readiness against the UTC Mission Capability (MISCAP) and ensure mission reliability. UTC PEs are completed on all 1C8 personnel assigned to UTC-based organizations whose primary mission is to deploy. An individual will complete these evaluations within 90 calendar days (180 calendar days for ANG/AFR) of being identified in the appropriate readiness tracking database. **(T-3).** These evaluations can be accomplished during field training exercises and other internal training opportunities. Team UTC PEs are encouraged; however, these evaluations do not apply to UTCs that do not have an equipment/technical component. **(T-3). Note:** This evaluation can replace or be accomplished in conjunction with the initial IPE. If an individual is permanently reassigned to another UTC or the primarily assigned UTC has major changes (e.g., equipment replacement, new tasks identified) another UTC PE is required.

7.4.2.3. **Special Personnel Evaluation.** SPEs are conducted on an as-needed basis. Commonly these evaluations are conducted after major modifications of equipment/systems, or reevaluation of an individual who received a Not In Compliance (NIC) result from a previous evaluation, or who has had his/her qualification status revoked for any reason other than supervisor decertification.

7.4.2.3.1. Modification SPEs are required within 60 calendar days of new equipment/system acceptance on a sampling of assigned Regular Air Force personnel. **(T-3).** Modification SPEs are required within 120 calendar days of new equipment/system acceptance on a sampling of ANG and AFR personnel. **(T-3).** A major modification is defined as changes to published operating system parameters, specifications, requirements or any change that revises published safety procedures.

7.4.2.3.2. Reevaluation SPEs are the assessment of an individual who received a NIC result from a previous evaluation, or who has had his/her qualification status revoked for any reason other than supervisor decertification. Reevaluation will be conducted within 90 calendar days after notification of recertification by the individual's supervisor. **(T-3).** ANG and AFR will be reevaluated within 120 calendar days. **(T-3).** The reevaluation of an individual whose previous evaluation resulted in a NIC will include only the specific tasks graded NIC, or the combination of the tasks that resulted in the NIC, unless the individual's leadership determines a complete reevaluation is necessary. **(T-3).**

7.4.2.4. During the Unit Effectiveness Inspection (UEI) Capstone event, the respective IG team will validate and verify the results of the I&E program for all applicable PEs. **(T-3).**

7.4.3. **Conducting Evaluations.**

7.4.3.1. **Evaluators.** Evaluators will be a MSgt, work center supervisor, 7-level technician, or civilian equivalent**. (T-3)**. The evaluator will not be the individual's trainer. **(T-3).** Evaluators must make careful observations of actions taken to accomplish the process being evaluated and each evaluator shares the duty for safe mission conduct with the individual being evaluated. **(T-3).** Whenever an evaluator observes a breach of security, safety, discipline, or procedure during the evaluation, the evaluator will take immediate corrective action to ensure mission operations, security, and safety. **(T-3).** Evaluators can serve as safety observers if they meet all safety requirements.

7.4.3.2. **Before the Evaluation.** Before the evaluation, evaluators must analyze and select tasks to be evaluated based on the type of evaluation, mission requirement, trend analysis indicators, training management visits, system performance data (metrics), previous evaluations done in the work center, and other management indicators requiring special emphasis. **(T-3).** Evaluators then select tasks not previously evaluated in the work center of which at least two-thirds are performance-based and the remaining third are common knowledge core tasks as identified in the Individual Training Plan (ITP). For UTC PEs, the evaluator will assess a minimum number of tasks to ensure proper pre-deployment, employment, and redeployment of the assigned UTC. **(T-3).** These tasks will be selected from the individual's ITP and align to the UTC MISCAP. **(T-3).**

7.4.3.2.1. Consider all equipment when selecting tasks for PE to adequately assess the work center training program effectiveness.

7.4.3.2.2. Select alternate tasks to avoid the need to reschedule an evaluation when operational requirements do not permit completion of planned evaluations.

7.4.3.2.3. Evaluators shall brief the technician on the tasks to be evaluated, the rating/grading criteria, and the performance standards prior to beginning the evaluation. **(T-3).** Evaluators focus on the evaluation process, not the specific evaluation steps the individual must accomplish to pass. Emphasize that the goal is to determine the effectiveness of the work center's training program, not to pass the evaluation. The evaluator must advise the technician of the following:

7.4.3.2.3.1. The evaluation starts when the technician begins the task, or portion of the task to be evaluated, and is completed when the entire task or previously determined portion of the task is finished. **(T-3).**

7.4.3.2.3.2. All maintenance actions performed are subject to evaluation. **(T-3).**

7.4.3.2.3.3. Safety, security, or system/equipment integrity will not be compromised for any reason. **(T-3).**

7.4.3.2.3.4. All detected errors during the evaluation will be used to calculate an overall award of Complies (C), Complies With Comments (CWC), or NIC. **(T-3).**

7.4.3.2.3.5. The technician must notify the evaluator of applicable information that could affect the task evaluation. **(T-3).** This includes any Previously Complied With (PCW) task(s)/step(s) (e.g. a setup task for tactical equipment requires

driving a ground rod, however the evaluation is conducted in a bay, so the technician hooks to a facility ground instead). If the technician fails to do this, he or she may be charged with an error for requirements that were omitted during the task performance that were not identified as PCW prior to the evaluation critique. **(T-3).**

7.4.3.2.3.6. The evaluator may ask questions or inject a task to determine technician knowledge of the task under evaluation. The technician may use technical references to answer any questions.

7.4.3.3. **During the Evaluation.** Evaluators will evaluate the process from beginning to end. **(T-3).** The process refers to three phases: preparation, task performance, and post completion procedures. Reference: AFMQCC 100-1, *Personnel Evaluation Report*.

7.4.3.3.1. Evaluators will stop the evaluation if technicians use methods or procedures that could jeopardize safety, violate security, or cause system/equipment damage. **(T-3).** Task evaluations may be continued (at the evaluator's discretion) after the hazard is corrected.

7.4.3.3.2. Evaluators will ask questions on the system knowledge, methods, and procedures used by the technician. **(T-3).** However, questions should not distract the technician from the task at hand. Questioning before and after task accomplishment is best.

7.4.3.3.3. Evaluators will assess a technician's task performance in accordance with **Attachment 2** of this AFMAN. **(T-3).** Technicians shall accomplish the task independently and without outside assistance unless the task requires multiple personnel as directed by guidance. **(T-3).**

7.4.3.3.4. Evaluators will report TO errors or deficiencies discovered during evaluations in accordance with TO 00-5-1. **(T-3).** Evaluators will report deficiencies in AFMSIs in accordance with **Chapter 12** of this AFMAN. **(T-3).** The work center will correct deficiencies in LWC. **(T-3).**

7.4.3.3.5. The evaluator may terminate the remaining portion of a task if the technician has demonstrated technical competency. The evaluator shall not terminate all remaining tasks for the evaluation. **(T-3).**

7.4.3.3.6. Evaluators will capture specific examples of how the individual performed various tasks for the report. **(T-3).**

7.4.3.4. **After the Evaluation.** At the conclusion of the evaluation, evaluators will provide an initial out-brief to the work center supervisor and the technician. **(T-3).** Evaluators will complete and submit a written report in accordance with **Attachment 2**. **(T-3).**

7.5. Equipment Evaluation (EE).

7.5.1. The evaluation of systems maintained by the work center will be conducted by the I&E program on a 12-month cycle. **(T-3).** **Attachment 3**, *Sampling Plan*, can be utilized for EEs consisting of multiple like items. EEs will be conducted as follows:

7.5.1.1. EEs give an overall view of the quality of maintenance. They also provide useful data for identifying training deficiencies and potential problem areas. **(T-3).**

7.5.1.2. Evaluators will perform EEs in enough depth to ensure systems and equipment are maintained and managed according to applicable technical data. **(T-3).** The following areas will be evaluated as a minimum:

7.5.1.2.1. System performance as indicated by the technical parameters outlined in equipment specific TO/TM guidance. **(T-3).** Also, NAVAIDS and ATC radar systems performance will be compared to flight inspection records and facility/equipment reference data. **(T-3).**

7.5.1.2.2. Equipment cleanliness and housekeeping. **(T-3).**

7.5.1.2.3. Compliance with standard maintenance practices and equipment safety requirements. **(T-3).**

7.5.1.2.4. Corrosion prevention and control. **(T-3).**

7.5.1.2.5. Equipment historical files, TO/TM configuration, and completeness using equipment inventory AFTO Form 470, AFTO Form 471, AFTO Form 472, automated copies according to TO 00-35D-2, *Electronic Set Inventory Checklist for Ground Communications-Electronic (C-E) Equipment*, and/or master COTS inventories. **(T-3).** Units will develop master inventories based on shipping, acceptance, catalog, and other applicable documents for COTS equipment. **(T-3).**

7.5.1.2.6. Applicable mobility markings. **(T-3).**

7.5.1.2.7. Compliance with preplanned and time change requirements. **(T-3).**

7.5.1.2.8. Management action taken on deferred maintenance actions. **(T-3).**

7.5.1.2.9. Verification of the Master PMI Schedule to ensure all PMIs are scheduled. **(T-3).**

7.5.1.2.10. Availability and condition of the required technical data, tools, and TMDE. **(T-3).**

7.5.1.3. Use of appropriate AFMQCCs is mandatory for all technical evaluations. **(T-3).** For equipment where AFMQCCs have not yet been published, local check sheets may be used.

7.5.1.4. Do not disassemble equipment solely for evaluation purposes. **(T-3).** However, if disassembly is required to conduct the evaluation, make every effort to perform the evaluation in conjunction with a frequent interval PMI, maintenance action, or inspection. **(T-3).**

7.5.1.5. Evaluators will perform EEs on a sample of all equipment on a 12-month basis using the sampling plan in **Attachment 3**. **(T-3). Note**: The commander may increase the frequency of technical evaluations based on mission requirements, suspected training problems, uptime rates, etc. AFR and ANG will complete technical evaluations on a 24-month basis. **(T-3).**

7.5.1.6. EEs are most effective if performed shortly after scheduled maintenance is performed. Perform these evaluations in conjunction with a short interval PMI or other maintenance action to minimize operational mission disruption. **(T-3).**

7.5.1.7. At the conclusion of the evaluation, evaluators will provide an initial out brief to the work center supervisor. **(T-3).** Evaluators will complete and submit a written report in accordance with **Attachment 2**. **(T-3).**

7.5.1.8. The most recent EE report will be maintained in the equipment historical record. **(T-3).**

7.5.2. During the UEI Capstone event, the respective IG team will validate and verify the results of the I&E program for all applicable EEs. **(T-3).**

7.6. Unit Type Code Evaluations.

7.6.1. UTC evaluations will consist of three separate components designed to measure readiness: personnel readiness, equipment readiness, and acceptance readiness when new or overhauled equipment is received from the manufacturer/depot. **(T-3).** The components of the UTC inspection can be accomplished separately or grouped together into one evaluation depending on availability of equipment and personnel. If grouped together, generate one report versus separate reports for each equipment end item/person. **(T-3).**

7.6.2. The evaluation intervals are as follows:

7.6.2.1. UTC personnel will be evaluated when assigned to a UTC or once every 24 months if already assigned to a UTC. **(T-3).** Number of tasks evaluated is at the discretion of the evaluator but should be UTC readiness oriented.

7.6.2.2. UTC equipment shall be evaluated as follows: pre-deployment, post deployment, and once every 24 months beginning from last post deployment inspection if not deployed. **(T-3).**

7.6.2.3. UTC acceptance evaluations shall be conducted as required when equipment is received from the manufacturer or depot. **(T-3).**

7.6.3. UTC PEs will be conducted at the same interval as standard PEs. **(T-3).**

7.6.4. UTC EEs assess mission readiness against the UTC MISCAP and ensures mission reliability. UTC EEs are completed on all equipment assigned to UTC-based organizations whose primary mission is to deploy. These evaluations are intended to ensure that equipment is operating within the same capabilities as reported in Status of Reporting and Training System (SORTS)/ Air and Space Expeditionary Force Reporting Tool (ART). The evaluations can be accomplished during field training exercises and other internal training opportunities when available. Team UTC EEs are encouraged. These evaluations do not apply to UTCs that do not have an equipment/technical component.

7.6.4.1. UTC EEs should include the highest interval PMI and below to evaluate that equipment is operating within tolerances in accordance with governing TOs/TIs/TMs and/or COTS documentation.

7.6.4.2. Inspections should also evaluate that known equipment limitations or malfunctions at the time of inspection are being accurately reported in the IMDS and/or required reporting tool.

7.6.4.3. A qualified 5-level or higher must conduct the UTC evaluation. **(T-3).** After completion, the UTC Team Lead and Flight Commander must review the results. **(T-3).**

7.6.5. UTC Acceptance Inspections (AI) will ensure that system(s) being accepted is/are within full operational capability in accordance with applicable TOs/TIs/TMs and/or COTS documentation and documented on applicable acceptance paperwork. **(T-3).** UTC AIs will also include completing a full inventory of all equipment supplied by the manufacturer or depot. **(T-3).** These inventories will be completed on AFTO Form 470, AFTO Form 471, and AFTO Form 472. **(T-3).**

7.6.6. During the UEI Capstone event, the respective IG team will validate and verify the results of the I&E program for all applicable UTC evaluations. **(T-3).**

Chapter 8

WORK CENTER SAFETY

8.1. Work center safety functional areas.

8.1.1. Work center supervisor will maintain a work center safety program. **(T-3).**

8.1.2. AFMAN 91-203, is the primary safety guidance for Air Force personnel.

8.1.3. AFI 91-202 outlines other safety program requirements and management information.

8.2. Work Center Safety Program. As a minimum, the work center safety program will contain:

8.2.1. **Lockout/Tag Out.**

8.2.1.1. LOTO will be accomplished in accordance with system TO requirements as outlined in AFMAN 91-203. **(T-3).**

8.2.1.2. In the event a completed Job Safety Analysis or Bioenvironmental Engineering's Occupational and Environmental Health risk assessment survey has determined the need for a more extensive LOTO program, those guidelines must be followed. **(T-3).**

8.2.2. **Ground and Lightning Protection/Electrostatic Discharge.**

8.2.2.1. Grounding, bonding, and shielding are approached from a total system concept by the DoD. TO 31Z3-822-2; TO 31-10-24-1, *Installation Practices: Communication Systems Grounding, Bonding and Shielding*; MIL-STD-188-124, *Grounding, Bonding and Shielding for Common Long Haul/Tactical Communication Systems Including Ground Based Communications-Electronics Facilities and Equipment* and MIL-HDBK-419A, *Grounding, Bonding, and Shielding for Electronic Equipment and Facilities*, layout the requirements of the subsystems.

8.2.2.2. Lightning Protection Subsystem consists of a network of conductors which provide a nondestructive path to earth ground (Earth Electrode Subsystem) for lightning energy. Reference MIL-STD-188-124 for an explanation of lightning and its effects.

8.2.2.3. Inspection Requirements. Ensure Base Civil Engineer (BCE) performs facility grounding and lightning protection checks in accordance with AFI 32-1065, *Grounding Systems.* **(T-3).** Ensure work center facility managers and safety monitors perform physical/visual grounding and lightning protection inspections as part of PMIs and required site inspections. **(T-3).** This includes mobile/transportable systems when operated at their garrison location.

8.2.2.4. Refer to TO 00-25-234 for ESD requirements.

8.2.3. **Climbing.**

8.2.3.1. Work centers will adhere to program requirements in accordance with AFMAN 91-203. **(T-0).**

8.2.3.2. Line worker boots must have safety toes that meet ASTM F2412-05, *Standard Test Methods for Foot Protection,* standards. **(T-0).**

8.2.4. **Arc Flash.**

8.2.4.1. Work centers are required to adhere to safety requirements outlined in National Fire Protection Association (NFPA) 70E, *Standard for Electrical Safety in the Workplace*. **(T-0).**

8.2.4.2. Work centers will provide and maintain Arc Flash protective gear to personnel as outlined in NFPA 70E. **(T-0).**

8.2.5. Work center will adhere to individual MAJCOM, FOA, or DRU supplements for AFMAN 91-203 and AFI 91-202 that may levy additional safety requirements.

Chapter 9

TRAINING MANAGEMENT

9.1. Introduction.

9.1.1. The Air Force OJT Program provides training for personnel to attain knowledge and skill qualifications required to perform duty in their specialty.

9.1.2. All training will be conducted in accordance with AFI 36-2670, and AFI 36-401, *Civilian Training, Education, and Professional Development*. **(T-3).**

9.2. Training Program Requirements.

9.2.1. Work Center training requirements are available in AFI 36-2670.

9.2.2. All 1C8 personnel training will be tracked and documented in the TBA except for ancillary training. **(T-3).**

9.2.3. All Air Force required ancillary training will be tracked in Advanced Distributed Learning Service unless otherwise specified by program guidance. **(T-3).**

9.2.4. All non- Air Force required ancillary training will be tracked in an Air Force-approved AIS such as IMDS. **(T-3).**

9.3. Climbing Training Requirements.

9.3.1. Climbing training guidance and requirements are available in AFMAN 91-203.

9.3.2. Technicians will receive initial training from a qualified training certifier at locations where climbing is required. **(T-1). Note:** This is directed towards technicians that did not receive initial climbing training in an apprentice course. Technicians will be certified on structures according to task requirements of the work center they are assigned. **(T-1).**

9.3.3. Installations without a climbing certifier may use an alternate source for climbing certification (e.g., Air Force mobile training team, local climbing courses, E&I, cable and antenna teams).

Chapter 10

LOGISTICS, LIFE CYCLE, AND PROJECT MANAGEMENT

10.1. Introduction.

10.1.1. This chapter provides logistics and life cycle management procedures for mission systems and equipment.

10.1.2. It is intended to assist and direct managers in a systematic approach to resolving mission systems and equipment sustainability problems and making critical logistics decisions.

10.1.3. This chapter provides framework and elements to establish sustainability processes for mission systems and equipment support.

10.2. Air Force Centrally Supported Equipment/Systems.

10.2.1. Centrally supported equipment/systems are managed by a Program Office in accordance with AFI 63-101/20-101.

10.2.2. A Program Office will be responsible for the equipment's logistics life cycle management and Operational Safety, Suitability, and Effectiveness (OSS&E) in accordance with Air Force Materiel Command Instruction (AFMCI) 63-1201, *Implementing Operational Safety, Suitability and Effectiveness (OSS&E) and Life Cycle Systems Engineering (LCSE)*. **(T-3).**

10.3. Non-Centrally Supported Equipment/Systems.

10.3.1. A Program Office does not manage non-centrally supported equipment. For this type of equipment, the procuring activity (e.g., MAJCOM, FOA, DRU, or unit) will be responsible for all life cycle sustainment planning and OSS&E. **(T-3).**

10.3.2. Effective logistics life cycle management for non-centrally supported equipment is essential to meet mission requirements.

10.4. Life Cycle Management Plans.

10.4.1. Whether purchased for Air Force, MAJCOM, FOA, DRU, base or unit use, all mission systems and equipment will have LCMP. **(T-3).** MAJCOM/FOA/DRU /Base/Unit acquired COTS/GOTS systems and equipment will document the sustainment support strategy in an LCMP. **(T-3).** The LCMP starts from the high-level logistics support strategy and becomes more refined. By the Production and Deployment phase (Joint Capabilities Integration & Development System - Milestone C) or Build and Test phase (Information Technology (IT) Lean Reengineering - Test Readiness Review 2), the LCMP will be very detailed.

10.4.2. In accordance with Air Force Pamphlet (AFPAM) 63-128, *Integrated Life Cycle Management,* as a minimum, managers must consider the following LCMP sustainment elements to help manage, control, sustain, and dispose of mission systems and equipment:

10.4.2.1. Sustainment/Systems Engineering. **(T-3).**

10.4.2.2. Design Interface. **(T-3).**

10.4.2.3. Supply Support. **(T-3).**

10.4.2.4. Maintenance Planning and Management. **(T-3).**

10.4.2.5. Support Equipment/Automatic Test Systems. **(T-3).**

10.4.2.6. Facilities. **(T-3).**

10.4.2.7. Package, Handling, Storage, and Transportation. **(T-3).**

10.4.2.8. Technical Data Management/TOs. **(T-3).**

10.4.2.9. Manpower and Personnel. **(T-3).**

10.4.2.10. Training. **(T-3).**

10.4.2.11. Computer Resources Support. **(T-3).**

10.4.2.12. Protection of Critical Program Information and Anti-Tamper Provisions. **(T-3).**

10.5. MAJCOM, Field Operating Agency, Direct Reporting Unit Functions. MAJCOM, FOA, or DRU s must provide adequate logistics support availability for sustained operations for MAJCOM, FOA, or DRU -acquired COTS/GOTS systems and equipment. MAJCOM, FOA, or DRU s will:

10.5.1. Develop approved procedures necessary to procure MAJCOM/FOA/DRU /base/unit systems/equipment.

10.5.2. Develop a LCMP for MAJCOM, FOA, or DRU -acquired COTS/GOTS systems/equipment early in the acquisition process (**See Chapter 13**).

10.5.3. Use the LCMP and budget appropriately for the acquisition and logistics support through all phases of the system/equipment life cycle (concept through disposal).

10.5.4. Provide unit guidance and support concerning system/equipment analysis, reliability, availability, and maintainability programs. The Standard Reporting Designator (SRD) assignment will be provided by MFMs for MAJCOM, FOA, or DRU unique systems or authorize the use of an approved contractor management information system with system/equipment control metrics.

10.5.5. Perform trend analysis and conduct special studies, when determined necessary, to identify adverse equipment performance and provide feedback to units.

10.5.6. Make units aware of Contracts and Acquisition and logistics sustainment requirements. Other key knowledge references include: AFI 10-601, *Operational Capability Requirements Development*; AFI 63-101/20-101; AFPAM 63-128; the Defense Acquisition Guidebook at **https://dag.dau.mil/Pages/Default.aspx**. Details are also available on the Integrated Defense Acquisition, Technology, and Logistics Life Cycle Management Framework link at **https://dag.dau.mil/Pages/acqframework.aspx**.

10.6. Base/Unit Functions. Bases/units must ensure logistics support is available for sustained operations for unit-acquired COTS/GOTS systems/equipment. **(T-3).** Bases/units will:

10.6.1. Use referenced publications, forms, and MAJCOM, FOA, or DRU -approved procedures to procure base/unit level systems/equipment. **(T-3).**

10.6.2. Develop an LCMP for Base/Unit-acquired COTS/GOTS systems/equipment early in the acquisition process (**See Chapter 13**). **(T-3).**

10.6.3. Use the LCMP and budget appropriately for the acquisition and logistics support through all phases of the system/equipment life cycle. **(T-3).**

10.6.4. Take prompt action to resolve logistics support problems and request assistance through appropriate channels when necessary. **(T-3).**

10.7. Work Center Supervisor.

10.7.1. Supervisors shall ensure work center logistics support management and work center project coordinator duties are accomplished and appropriately documented. **(T-3).**

10.7.2. Supervisors shall ensure support requirements for new systems, programs, and plans are established. **(T-3).**

10.7.3. Supervisors must also assist in the preparation of the maintenance budget estimate. **(T-3).**

10.7.4. Supervisors must understand all aspects of work center logistics support programs. **(T-3).** Supervisors must:

10.7.4.1. Appoint a work center project manager for each E&I, contractor, or self-help project to ensure project manager duties are accomplished. **(T-3).**

10.7.4.2. Assign one or more technicians to work with E&I project and special maintenance teams. **(T-3).** The maintenance work center's commander/superintendent may waive this requirement on a case-by-case basis. However, assigned technicians receive valuable training from the team and can train other work center technicians after the team departs.

10.7.4.3. Work with the MFM and the maintenance work center's commander/superintendent to resolve issues concerning manpower authorizations and allocations for the work center. **(T-3).**

10.7.4.4. Review new work center extended Unit Manpower Document when issued. **(T-3).**

10.7.4.5. Help prepare work center manpower change requests. **(T-3).**

10.7.4.6. Help prepare work center manpower standard applications or reapplication. **(T-3).** Supervisor must be familiar with the work center's manpower standards. **(T-3).** See AFPAM 63-128 for specific functions the work center supervisor may need to perform in support of the PM.

10.7.4.7. Provide budget estimate inputs to the commander/superintendent and monitor work center expenditures. **(T-3).**

10.7.4.8. Develop annual budget estimates and amended estimates. **(T-3).**

10.7.4.9. Include support for programmed systems and equipment in budget estimates. **(T-3).**

10.7.4.10. Review support agreements to identify special support requirements. **(T-3).** Identify training, equipment, supply, vehicles, additional manpower, or other key elements required to provide the support. **(T-3).**

10.7.5. Work center project coordinators act as the work center focal point for all matters concerning the assigned project. Project Coordinators will:

10.7.5.1. Ensure projects are accomplished with minimum difficulty and the work center can support systems or equipment programmed for installation or major modification. **(T-3).**

10.7.5.2. Work closely with the Program Office, E&I engineers, and E&I teams. **(T-3).**

10.7.5.3. Participate in site surveys and provide technical advice to the E&I or contractor team, work center, commander/superintendent, and the Program Office. **(T-3).**

10.7.5.4. Coordinate with other work center project coordinators to identify and resolve conflicts (such as storage space, power requirements or programmed equipment locations). **(T-3).**

10.7.5.5. Provide continuity of logistics support preparations for the project. **(T-3).**

10.7.5.6. Review E&I project packages and amendments. **(T-3).**

10.7.5.7. Initiate engineering change requests in accordance with TO 00-33D-2002, *Methods and Procedures for Cyberspace Engineering Installation Activities Management*, for necessary changes to the project for deficiencies noted in the project package. **(T-3).**

10.7.5.8. Identify changes and deficiencies before the installation begins to ensure timely project completion and to prevent delays and work stoppages during installation. **(T-3).**

10.7.5.9. Identify all enroute, field training documents, and other formal training required to support programmed systems and equipment. **(T-3).**

10.7.5.10. Coordinate with the E&I project team to obtain required technical data, test equipment, special tools, other support equipment, training, manpower, and so forth. **(T-3).**

10.7.5.11. Coordinate with the work center test equipment monitor to ensure calibration capability is established for new test equipment authorized for programmed systems and equipment. **(T-3).**

10.7.5.12. Appoint technicians to work with the E&I project team, monitor progress of the project, and participate in systems or equipment AIs. **(T-3).**

10.7.5.13. Use Cyberspace Infrastructure Planning System (CIPS) to document project actions per TO 00-33D-3005, *Methods and Procedures - Managing the Cyberspace Infrastructure with the Cyberspace Infrastructure Planning Systems (CIPS).* **(T-3).**

10.8. Modification Management.

10.8.1. A modification proposal is a recommendation to change the operation, use, or appearance of Air Force centrally supported equipment/systems in accordance with AFI 63-101/20-101. Only MAJCOM, FOA, or DRU/OSS&E authority may approve modifications, whether temporary, permanent, or for safety, to mission equipment or systems. **(T-3).** Submit modification requests via change management process. **(T-3).**

10.8.2. AF Form 1067, *Modification Proposal*, *Process*. The AF Form 1067 is required for a specific improvement to maintenance or operation. The process begins with the initiator, who

fills out the AF Form 1067, gains base approval, and then sends to the MAJCOM, FOA, or DRU for initial validation. The form and request are then routed through the coordination chain.

10.8.2.1. Upon MAJCOM, FOA, or DRU approval, the AF Form 1067 request will be routed to the appropriate Lead Command.

10.8.2.2. Lead Commands will validate the request and route to the appropriate Program Manager (PM).

10.8.2.3. The PM, with assistance of equipment specialists and engineering, will review the technical validity and OSS&E impact of the request and make a determination to implement or deny the request.

Chapter 11

MATERIEL MANAGEMENT

11.1. Introduction. Materiel management is critical for mission accomplishment. Every person is responsible to manage and control all assets in the most cost effective and efficient manner.

11.1.1. Supply discipline is everyone's responsibility. Supply discipline means to do everything within one's power to do the job with minimum cost to the Air Force.

11.1.2. Supply discipline/materiel management not only refers to not spending money when it is not needed, but also making sure equipment and supplies are properly accounted for, protected from loss or damage, used for their intended purpose, and returned to the supply system or disposed of properly when no longer needed. This includes sending items to Defense Logistics Agency Disposition Services.

11.2. Logistics Readiness Squadron Liaison. The LRS liaison coordinates with support agencies and assists work center personnel by expediting all supply transactions. The LRS liaison is the primary liaison between the work center and LRS/supply activities. Refer to AFMAN 23-122, *Materiel Management Procedures*, and TO 00-20-10.

11.2.1. The LRS liaison duties may be completed as an additional duty by work center personnel if the unit does not have a designated liaison.

11.2.2. The LRS liaison will:

11.2.2.1. Complete all LRS required training to perform these duties.

11.2.2.2. Coordinate with the LRS to set up organizational codes, shop codes, delivery destinations for property (e.g., issues and due-out releases), and to process serviceable and unserviceable turn-ins. **(T-3).**

11.2.2.3. Maintain close relations with the LRS and/or the applicable Logistics Support Center to ensure Mission Capable (MICAP)-reportable items are reported and NMC supply requirements are satisfied. **(T-3).** Ensure correct use of Force Activity Designator, Urgency of Need Designator (UND) and requisition priorities. **(T-1).** Be familiar with LRS after-hours procedures. **(T-3).**

11.2.2.4. Set up procedures to route, store, and control repair cycle assets and act as the repair cycle monitor to include Due-In-From Maintenance (DIFM), Awaiting Parts (AWP), and Equipment Inoperative for Parts for the maintenance work center. **(T-3).** Ensure proper management of repair cycle assets. **(T-3).**

11.2.2.5. Act as the monitor for the work center Low Density Level requests. **(T-3).** The request will be completed using format designated and submitted through the local supply customer service. **(T-3).**

11.2.2.6. Coordinate with the LRS to ensure all required TCTO kits and Time Change Item (TCI) are promptly requisitioned and delivered to the maintenance work center. **(T-3).**

11.2.2.7. Coordinate with the LRS to ensure TCTO actions are accomplished on supply-controlled spares. **(T-3).**

11.2.2.8. Serve as Turn-Around (TRN) monitor **(T-3).**

11.2.2.9. Advise the maintenance work center on the use of repair cycle asset initial issue procedures. **(T-3).**

11.3. General Materiel Management.

11.3.1. Supervisors ensure cost-effective mission support through the proper use and management of supply assets, support equipment, and local purchase materials.

11.3.2. Supervisor will:

11.3.2.1. Comply with the Precious Metals Recovery program in accordance with AFI 23-101. **(T-3).**

11.3.2.2. Use appropriate AIS to requisition parts directly whenever possible. **(T-1).**

11.3.2.3. Use the direct call-in method between the work center and LRS customer service/demand processing when IMDS or Integrated Logistics Supply System (ILS-S) is not available. **(T-3).**

11.3.2.4. Use AF Form 2005, *Issue/Turn-In Request*, or any other LRS approved control register to document requests for direct demands on supply. **(T-3).** Automated supply log from IMDS is authorized. Verify UND "A" and "B" requests prior to call-in. **(T-3).**

11.3.2.5. Properly manage repair cycle assets. **(T-3).**

11.3.2.6. Notify the DIFM monitor of status changes for assets kept in the work center. **(T-3).**

11.3.2.7. Process repaired assets and review associated NRTS paperwork in accordance with TO 00-20-10. **(T-3).**

11.3.2.8. Process repairable property under warranty or guarantee in accordance with TO 00-20-10. **(T-3).**

11.3.2.9. Submit TRNs in accordance with TO 00-20-10 and AFMAN 23-122. **(T-3).**

11.3.2.10. Coordinate with LRS to ensure RSP assets requiring functional checks are identified. **(T-3).** Notify LRS when functional checks are completed. **(T-3).**

11.3.2.11. Submit deficiency reports or reports of discrepancy when deficient materiel is received in accordance with local procedures or follow the manufacturer's instruction. **(T-3).**

11.3.2.12. Monitor and control bench stock in accordance with AFMAN 23-122. **(T-3).**

11.3.2.13. Review applicable Allowance Standards (AS) to identify additions, deletions, and changes to work center support equipment authorizations. **(T-3).** Perform AS review biennially. **(T-3).** Submit recommended changes in accordance with AFMAN 23-122. **(T-3).**

11.3.2.14. Verify TCTO kits are correct and complete when received from the LRS. **(T-3).**

11.3.2.15. Notify equipment custodians when TCTO actions result in equipment stock number changes as applicable. **(T-3).**

11.3.2.16. Identify preplanned items, TCI, and TCTOs. **(T-3).** Complete required actions when scheduled. **(T-3).**

11.3.2.17. When authorized by the commander/superintendent **(**or applicable level in accordance with local organizational structure), establish and manage work order residue. **(T-3).** Maintain a list of items on work order residue and develop procedures to encourage consumption of work order residue prior to using bench stock. **(T-3).**

11.3.2.18. Develop written guidance to monitor and control shop/operating stocks in accordance with AFI 23-111. **(T-3).** For explanatory purposes, shop/operating stocks are those items (e.g., General Services Administration (GSA) purchased cable stocks, connectors, hardware) purchased with Air Force funds to fulfill mission requirements (e.g., PMIs, equipment maintenance, jobs) that cannot be loaded on bench stock or other accounts. **Note:** Work center supervisors should establish procedures covering part number, purchasing information, condition, etc.

11.3.2.19. Verify items critical to deploying equipment are stocked, maintained, and secured. **(T-3).** These items will not be used for in-garrison requirements. **(T-3).**

11.3.2.20. Verify the equipment custodian completes and updates the applicable report to address incoming or outgoing equipment/systems when E&I, self-help, installation, or removal projects are complete. **(T-3).**

11.3.2.21. Manage forward supply point assets in accordance with AFMAN 23-122. **(T-3).**

11.3.2.22. Comply with ESD storage procedures in accordance with AFI 23-101. **(T-3).**

11.3.2.23. For Air Force -supplied/depot-supported systems, ensure that all spare assets supplied during installation or modification are brought to record and added to forward supply point. **(T-3).**

11.3.2.24. Return all useful consumable and non-consumable materials not identified as shop stock/work order residue to the LRS/Materiel Management Activity in accordance with AFMAN 23-122. **(T-3).**

11.3.2.25. Coordinate with the local LRS, Equipment Accountability Officer (EAO), and work center personnel to identify deploying assets. **(T-3).**

11.3.2.26. Identify all deploying assets to the applicable base-level Equipment Control Officer (ECO) upon receipt of mission tasking order during the pre-deployment actions. **(T-3).** For supply assets, follow the LRS procedures. **(T-3).**

11.3.2.27. Verify lists of deploying assets accompany the UTCs and the deploying equipment custodian to the Area of Responsibility (AOR). **(T-3).** Start asset transfers as soon as possible depending on the AOR policy and mission-tasking instructions. **(T-3).** Custodians will complete the required training prior to deploying. **(T-3).**

11.3.2.28. Upon return to home station, verify assets that were returned to the LRS, EAO, and ECO. **(T-3).**

11.3.2.29. Dispose and demilitarize designated materials in accordance with AFMAN 23-122, and Air Force Handbook (AFH) 23-123, Volume 1, *Materiel Management Reference Information.* **(T-3).**

11.3.2.30. Authorize and manage equipment per AFH 23-123, Volume 3, *Air Force Equipment Management.* **(T-3).**

11.3.2.31. **(Added-AETC)** Provide a listing of all AO mission support structures to the AETC RAWS staff. **(T-3)**

Chapter 12

TECHNICAL ORDER MANAGEMENT

12.1. Introduction.

12.1.1. AFIs, AFMANs, and technical publications are essential for maintenance organizations to function properly and to provide the maintenance activity with accurate information.

12.1.2. Technical publications include TIs/TOs/TMs and other specialized publications. Set up and maintain these publications in accordance with AFI 33-360 and TO 00-5 series publications.

12.2. Work Center/Section/Unit Type Code Publication Management.

A supervisor must:

12.2.1. Provide availability and strictly enforce adherence to and compliance with instructions, technical publications, and supplements. **(T-3).**

12.2.2. Establish procedures for shipping instructions, technical publications, and supplements to support mobility requirements. **(T-3).**

12.3. Technical Order Distribution and Control.

12.3.1. This section pertains to necessary actions for the distribution and control of TOs within the Air Force. **Note:** See TO 00-5-1 for Lead TODO functions.

12.3.2. The TODO responsibilities are to verify the adequacy and accuracy of TO files in the maintenance activity and will manage responsibilities in accordance with TO 00-5-1. **(T-3).** TODOs will:

12.3.2.1. Provide special attention to all electronically transmitted TCTOs and TOs due to the urgent nature of this type of change. **(T-3).**

12.3.2.2. Provide current methods and procedures TOs, TCTOs, evaluation work cards, work unit code (WUC) manuals, and other TMs to the entire maintenance activity. **(T-3).** The primary consideration is availability of TOs, with minimum duplication. **(T-3).** For more information on each of these publications, review TO 00-5-1. **(T-3).**

12.3.2.3. Properly manage all Electronic Tools (E-Tool) computers and work closely with network representatives to ensure hardware remains fully operational and network requirements are fully met to support real-time updates and the most current status of technical data and TCTOs as required by TO 00-5-1. **(T-3).** Refer to TO 31S5-4-ETool-1, *ETool and Commercial Mobile Device Setup and Management*, for guidance on set-up and management of E-Tools. **(T-3).**

12.3.2.4. Route equipment or policy TO waiver requests to MAJCOM, FOA, or DRU. **(T-3).**

12.4. Local Work Cards.

12.4.1. LWCs, Local Job Guides, Local Work Card Page Supplements, and Local Checklists are created by the using activity. PMI requirements are normally shown in work card sets, or maintenance TMs. Refer to TO 00-5-1 for guidance on the development and maintenance of LWC.

12.4.2. COTS items, especially electronic components, are bought and do not have PMIs published in the TO System. In this event, follow the manufacturer's recommended maintenance schedules provided in the technical guidance or complete PMIs per flight commander's/superintendent, or equivalent direction. LWCs may be developed.

12.4.3. The flight commander/superintendent, or equivalent, will determine if PMIs are to be accomplished in absence of commercial manuals or publications. **(T-3).** If there is not a flight commander/superintendent or equivalent, then this duty will be completed by the unit commander. **(T-3).** Refer to TO 00-5-1 and TO 00-20-1, *Aerospace Equipment Maintenance Inspection, Documentation, Policies, and Procedures*. Consider specifications for mean-time-between-failure, operational requirements, corrosion control, and other checks.

12.4.4. To report changes to technical data for centrally managed and procured equipment, AFTO Form 22, *Technical Manual (TM) Change Recommendation and Reply* or Enhanced Technical Information Management System (ETIMS) equivalent will be used. **(T-3).**

12.4.5. Non-AFMC-supported equipment/systems require technical data that must be provided by the Lead Command and applicable Program Management Office. **(T-3).** New systems will be fielded with all applicable data. **(T-3).**

12.5. Time Change Item Management.

12.5.1. TCI management includes such items as TCTOs and other Air Force/MAJCOM /FOA/DRU -directed modifications and inspections that provide units with instructions for completing a one-time change, modification or inspection of equipment.

12.5.2. All documentation directing modifications and inspections (with the exception of immediate and urgent action documentation) are considered scheduled maintenance and will be integrated into maintenance planning cycles. **(T-3).**

12.5.3. Consider concurrent accomplishment of modification and inspection work with other scheduled or unscheduled maintenance actions. When practicable, all Peacetime Operating Stock, spares, and RSP assets will be modified before use or installed in equipment. **(T-3).**

12.5.4. Time changes will be managed in accordance with TO 00-5-15 and specific MAJCOM, FOA, or DRU instructions. **(T-3).** Detailed guidance and codes for processing TCTOs are contained in Introduction to Integrated Maintenance Data System Central Database, and AFCSM 21-568, Volume 2.

12.5.5. For Program Office-managed systems with no SM support, refer to guidance for COTS/GOTS manuals/modifications in TO 00-5-1.

12.5.5.1. Organizations must establish an alternate method to process TCIs including TCTO data **(T-3).** TCTO acknowledgment forms will be completed per TCTO instructions and maintain copy of completed form. **(T-3).** Data will be entered into IMDS when restored for units experiencing a delay. **(T-3).**

12.5.5.2. Service bulletin requirements are addressed in TO 00-5-1 and TO 00-5-15.

12.5.6. The TODO will initiate TCTO processing actions in accordance with TO 00-5-15. **(T-3).**

12.5.6.1. Work center Technical Order Distribution Account (TODA) will review each incoming TCTO and advise the work center supervisor of its applicability. **(T-3). Note:** If the work center has a TODO that person may also fulfill TODA function.

12.5.6.2. For RMC-maintained systems, the RMC will coordinate with the RAWS work center to process, complete, and document completion of all TCTOs. **(T-3).**

12.5.6.3. For systems not maintained by the RMC, the work center will interface directly with unit TODO, determine applicability, complete and document TCTOs. **(T-3).**

12.5.7. Work centers will review all applicable -6 scheduled inspection and maintenance requirements manuals and -6WC work cards to determine if preplanned and TCIs are required. **(T-3).** When requirements exist, the work center will provide the appropriate authority the information needed to initiate suspense procedures. **(T-3).**

12.6. Federal Aviation Administration Technical Manuals Management.

12.6.1. The TODA will ensure FAA TIs, Joint Orders (JO), System Support Modifications (SSM), and System Documentation Releases (SDR) are current. **(T-3). Note:** If the work center has a TODO that person may also fulfill TODA functions.

12.6.2. All FAA documents should have a corresponding TO cover page in ETIMS. Users will be directed to the Aeronautical Data Exchange (ADX) Air Force community, **www.adx.faa.gov**.

12.6.3. FAA TIs and JOs shall be reviewed weekly in accordance with TO 00-5-1 to ensure all SDRs are being utilized. **(T-3).** If hard copy, TODO will document last review on front cover. **(T-3).**

12.6.4. SSMs are similar to TCTOs and are the method that the FAA utilizes to modify or update existing equipment. For Air Force operated systems, SSMs will be converted to a TCTO and distributed via ETIMS. **(T-3).** SSMs will not be performed until directed to do so via an Air Force TCTO. **(T-3).**

12.7. Air Force Maintenance Quality Control Checklist.

12.7.1. AFMQCCs are guides used to evaluate processes, equipment condition, maintenance quality, and maintenance management effectiveness. AFMQCCs are created using a standardized format. AFMQCCs and the standardized template can be accessed on the AFFSA SharePoint® page. See **Attachment 4** of this AFMAN for a listing of AFMQCC's and link to SharePoint®.

12.7.1.1. AFMQCCs for RAWS can be used in the absence of an Air Force-wide standardized checklist governing a particular process, function, or piece of equipment.

12.7.1.2. AFMQCCs do not prescribe nor establish Air Force-wide policy.

12.7.1.3. AFMQCCs must not be used to operate, maintain (e.g., tune, align, adjust) or troubleshoot equipment or processes. **(T-3).**

12.7.1.4. Equipment or management evaluations shall not be limited to the checks in the AFMQCCs. **(T-3).** MAJCOM, FOA, or DRUs are authorized to develop supplements to AFMQCCs, supplements to MQCCs, and LMQCC to check additional items based on operational needs. Forward proposed revisions of AFMQCCs to the checklist OPR.

12.7.1.5. Complete applicable AFMQCCs in accordance with **Attachment 3** of this AFMAN.

12.7.1.6. AFFSA centrally manages the RAWS AFMQCC program for standard processes and systems (e.g., used by more than one MAJCOM, FOA, or DRU). AFFSA maintains AFMQCC's through the following:

12.7.1.6.1. Maintaining a record copy of all published RAWS AFMQCCs.

12.7.1.6.2. Coordinating new AFMQCCs with appropriate MAJCOM, FOA, or DRUs to determine Air Force -wide applicability.

12.7.1.6.3. Assigning AFMQCC control numbers as appropriate and forward new AFMQCCs to the AFCFM for approval.

12.7.1.6.4. Publishing AFMQCCs and changes as required with approval from the AFCFM.

12.7.1.6.5. Assigning OCRs for AFMQCCs. **Note:** Lead Commands may designate OCRs, and MAJCOM OCRs may appoint unit level OCRs for appropriate AFMQCCs, but MAJCOM, FOA, or DRUs retain overall control for their respective AFMQCCs.

12.7.1.6.6. Coordinating annual reviews of AFMQCCs with appropriate OCRs for continued need, accuracy, and currency.

12.7.1.6.7. Make copies of all current AFMQCCs available on the AFFSA SharePoint®.

12.7.1.7. AFMQCC OPRs will:

12.7.1.7.1. Format AFMQCCs.

12.7.1.7.2. Review appropriate AFMQCCs annually for continued need, accuracy, and currency. Provide AFFSA/XA with changes and updates as required.

12.7.1.7.3. Evaluate and validate AFMQCCs before forwarding to AFFSA/XA for approval and publication.

12.7.1.7.4. Forward changes (except for minor punctuation and spelling errors) as AFMQCC revisions. To revise an AFMQCC, the complete AFMQCC is re-accomplished and processed in the same manner as new AFMQCCs

12.7.1.7.5. Send changes to AFFSA/XA for consolidation. Comply with MAJCOM, FOA, DRU, and/or local procedures for changing their AFMQCCs.

12.7.2. **Proposed Air Force Maintenance Quality Control Checklist.**

12.7.2.1. Accomplishing the evaluation and field-testing for proposed AFMQCCs by the lead MAJCOM, FOA, or DRU is required prior to forwarding them to AFFSA/XA.

12.7.2.2. Include the projected number of MAJCOM, FOA, or DRUs and users for proposed AFMQCCs.

12.7.3. **Local Maintenance Quality Control Checklist.**

12.7.3.1. Unit Commander will approve use and processing of LMQCCs but can be delegated no lower than the flight commander/chief. **(T-3).** Approved LMQCCs shall be forwarded to respective MAJCOM MFM/ FAM. **(T-3).**

12.7.3.2. LMQCCs will:

12.7.3.2.1. Be clearly marked or labeled as LMQCCs.

12.7.3.2.2. Not be retained or used after an Air Force, MAJCOM, FOA, or DRU MQCC is published on the same item of equipment, grouping of equipment (General AFMQCC) or management function.

12.7.3.2.3. Supplement Air Force, MAJCOM, FOA, or DRU MQCCs if local requirements dictate.

12.7.4. **Checklist Format.**

12.7.4.1. The title block of the AFMQCC contains the process, function, or nomenclature and noun of the equipment (normally the end item or system). If the title or type of equipment is not the end item, add the end item in parentheses (e.g., ASXXX antenna group (AN/GRT-XX Transmitter).

12.7.4.2. The "item" section of the AFMQCC will contain checks to determine overall performance, operation, and maintenance practices. Each item is limited to one sentence (if possible). Mark each item under appropriate column as either in compliance "Yes", not in compliance "No", or mark “N/A” if the question is not applicable to an item.

12.7.4.3. Do NOT duplicate individual checklist items currently published in any existing AFMQCC series checklists.

12.7.4.4. Do NOT use specific equipment parameter values (e.g., voltage and frequency measurements) in AFMQCCs.

12.7.4.5. Provide specific references (e.g., publication number and paragraph) for each equipment parameter and evaluation item.

12.7.4.6. Provide OPR name and office symbol.

12.8. Air Force Maintenance Special Instruction (AFMSIs).

12.8.1. This section establishes the procedures to develop, publish, distribute, implement, file, and dispose of a series of specialized publications for RAWS equipment. See **Attachment 5** of this AFMAN for a listing of AFMSIs.

12.8.1.1. AFMSIs are not published for use in place of Air Force TOs, FAA TIs, or JOs.

12.8.1.2. AFMSIs provide a means to issue inspection and servicing requirements, operational performance checks, and special maintenance instructions for which formal TO procedures are not published. They may also provide the means to issue optional or temporary modifications. AFMSIs will only be published for equipment that is applicable to more than one MAJCOM, FOA, or DRU. **(T-3).**

12.8.1.3. AFMSIs are published only after determining that the supporting Air Logistics Center (ALC) or contracted function cannot produce the required technical data or implement a permanent modification.

12.8.2. **Air Force Flight Standards Agency Operations, Training and Procedures Functions.**

12.8.2.1. Determine the needs, requirements, as well as process and publish proposed AFMSIs and changes.

12.8.2.2. Maintain record copies of AFMSIs.

12.8.2.3. Format modification-type AFMSIs as full-page documents. PMI-type procedures may be formatted as TO -6WC work cards (to fit six-hole binders) or as full-page documents, per OPR decision.

12.8.2.4. Submit proposed AFMSIs, recommended changes, additions or deletions through command channels to AFFSA in an electronic Word format. AFMSI OPRs will submit approved changes to AFFSA/XA.

12.8.2.4.1. AFMSIs are numbered in three series. Use the 100-series for instructions of a general nature, 200-series for special maintenance instructions and temporary modifications; and 300-series for inspection, servicing, lubrication requirements, and operational performance checks.

12.8.2.4.2. AFMSIs can be accessed on the AFFSA SharePoint® page. See **Attachment 5** of this AFMAN for a listing of AFMSIs and link to SharePoint®.

12.8.2.4.3. Implement AFMSIs on receipt unless otherwise directed. Schedule modifications, inspections, servicing, lubricating, and operational checks in the same manner as -6WC TOs/TCTOs.

12.8.2.5. Document accomplishment of AFMSIs as follows:

12.8.2.5.1. Scheduled Periodic Inspection, Servicing, Lubrication, and Operational Performance AFMSIs. Use the same procedures as those used to document -6WC TO work cards.

12.8.2.5.2. Special Maintenance Instructions/Temporary Modifications. Upon completion of the AFMSIs procedures, make appropriate entry in approved AIS historical record.

12.8.2.6. Rescind AFMSIs when no longer required or when material is incorporated into the TO. When an AFMSI should be rescinded, notify AFFSA/XA through command channels. **(T-3).**

Chapter 13

TMDE AND TOOLS MANAGEMENT

13.1. Introduction.

13.1.1. TMDE is critical to proper operation and maintenance of mission systems. Out-of-tolerance TMDE can cause mission systems to be misaligned or erroneously declared unserviceable. Man-hours and parts may be expended unnecessarily to restore already serviceable equipment. Personnel who use TMDE are required to understand their function and the procedures for the use and care of TMDE. Proper use, handling, storage, transportation, and calibration are essential to ensure TMDE accurately performs its function.

13.1.2. TMDE is defined as any device used to maintain, evaluate, measure, calibrate, diagnose or otherwise examine materials, supplies, equipment, and systems to identify or isolate actual or potential malfunction/discrepancies/deficiencies or decide if they meet operational specification established in technical publications. See TO 00-20-14 for details.

13.1.3. TMDE and Tool management applies to all sections supporting maintenance. All technicians will be knowledgeable of TO 00-25-234; TO 00-20-14; AFMAN 23-122; TO 33-1-27, *Logistic Support of Precision Measurement Equipment in FSC*; TO 33-1-32, *General Instructions for Input Power Wiring of Electrical/Electronic Support Equipment,* and any applicable commercial manuals/guides.

13.1.4. All work center technicians will complete and document AFJQS XXXXX-201P, *Work Center Test Equipment Management,* in accordance with AFI 36-2670. **(T-3).** If local tasks are required, ensure tasks are identified, added to the training plan, and training is conducted. **(T-3).**

13.1.5. Work centers will establish limited calibration requirements whenever possible. **(T-3).** This eliminates time spent to calibrate unused functions or ranges and expedites return of the TMDE to the work center. However, before establishing limited calibration, consider the following:

13.1.5.1. Determine the functions and ranges needed for all mission systems and equipment supported by the TMDE in the work center. **(T-3).** TMDE needs to be calibrated only for those functions and ranges used. **(T-3).**

13.1.5.2. TMDE shared by two or more work centers are required to be calibrated for the ranges and tolerances required by each using work center. **(T-3).**

13.2. General Test, Measurement and Diagnostic Equipment Requirements.

13.2.1. Priority calibration or repair will be requested through the work center TMDE monitor, only when justified to meet urgent mission requirements. **(T-3).** TMDE will be picked up from Precision Measurement Equipment Laboratory (PMEL) as soon as the priority calibration is completed. **(T-3).**

13.2.2. An extension of the calibration due date will be requested through the work center TMDE monitor, if loss of the TMDE delays or prevents critical mission accomplishment. **(T-3).**

13.2.3. Training will be provided or arranged for work center personnel on proper use and care of TMDE, including how to determine calibration conditions and limitations. **(T-3).**

13.2.4. The work center project coordinator must be advised of problems in obtaining or calibrating TMDE needed to install or maintain mission systems scheduled for installation. **(T-3).**

13.2.5. Commander or designated representative/equivalent must be notified when the lack of TMDE impacts completion of the work center's mission. **(T-3).**

13.2.5.1. Work centers must perform TMDE user applications in accordance with TO 00-20-14. **(T-3).**

13.2.5.2. Personnel are responsible to provide proper care, handling, cleanliness and transportation of TMDE in accordance with TO 00-20-14.

13.2.6. Replacement TMDE may be required if PMEL is unable to repair an item of TMDE, or when an item is condemned or designated as obsolete. When new or replacement TMDE is required, work center TMDE monitors will:

13.2.6.1. Determination of substitute items acceptability will be coordinated with the work center supervisor or 7-level. **(T-3).**

13.2.6.2. The work center supervisor and CA/CRL monitor will be notified when TMDE items are slated to be turned in to or received from supply. **(T-3).**

13.2.6.3. Obtaining applicable TOs for new TMDE and disposing of TOs for TMDE which has been turned in will be coordinated with the work center TO monitor. **(T-3).**

13.2.6.4. **(Added-AETC)** Replace TMDE items used as training support systems when they are no longer used by the career field. **(T-3)**

13.2.7. Recommend establishing TMDE inventory tracking system that facilitates quick inventory and accountability of TMDE assets.

13.3. General Tool Requirements.

13.3.1. Tool kits are defined as all government-purchased tools, to include tools provided by government-paid contractor, utilized to perform RAWS maintenance activities (restore/sustain/maintain systems). In this publication, the term "tool" refers to all types of tool kits as well as a basic tool (e.g., Xcelite, drill, and socket kits).

13.3.2. Tools are defined as are any instrument or apparatus used in performing an operation or task.

13.3.3. The objectives of the Tool Management Program are to reduce replacement costs through effective control and accountability of assets and minimize unsafe acts.

13.4. Work Center Tool Management.

13.4.1. This section pertains to the duties and functions at the work center level and applies to deployable tool kits as well as home station kits in accordance with TO 32-1-101-WA-1, *Use And Care Of Hand Tools And Measuring Tools*.

13.4.2. The tool program will address the following areas, as a minimum:

13.4.2.1. Appointment of a tool custodian and alternate in writing. **(T-3).** The unit may appoint a unit level tool custodian and alternate if desired instead of individual work center custodians.

13.4.2.2. Procedures addressing effective tool control and accountability. **(T-3).**

13.4.2.3. Procedures addressing tracking/using UTC/mobility tool kits. **(T-3).**

13.4.2.4. Managing industrial shop machinery tools and accessories/attachments (e.g., Hiltis, band saws, arbors, chucks, portable vise, blades, Pionjars) that normally do not fit within the tool kit. **(T-3).**

13.4.2.5. Establishing tool inventory tracking system that facilitates quick inventory and accountability of tools. **(T-3).**

13.4.2.6. Procedures addressing lost, broken or unserviceable tool replacement. **(T-3).**

13.4.2.7. Recurring inspection schedule for all tools and tool kits to ensure serviceability. **(T-3).**

13.4.2.7.1. Inventory tools, tool kits/boxes prior to first use of the day. **(T-3).** Inventory all used tools, tool kits/boxes again at the end of the shift. **(T-3).** If not used for the day, then no inventory is required. **(T-3).**

13.4.2.7.2. Inventory all tools, tool kits/boxes at least quarterly, regardless of usage. **(T-3).**

13.4.2.8. Procedures addressing management of tools with warranties and/or guarantees. Refer to AFMAN 17-1203, *Information Technology (IT) Asset Management (ITAM)*, for IT transactions; TO 00-20-3-WA-1, *Maintenance Processing of Repairable Property and Repair Cycle Asset Control System* for AFMC-managed systems. **(T-3).**

13.4.2.9. Procedures addressing control and management of replacement, expendable and consumable hand tools, and other items contained in tool kits or tools (e.g., connector pins, extraction pins, alcohol wipes). **(T-3).**

13.4.2.10. Procedures addressing accountability, control, and use of E-Tools. See AFMAN 17-1203 for further guidance. **(T-3).**

13.4.2.11. Procedures addressing positive control of rags when used in and around the flight line. **(T-3).**

13.4.2.12. Procedures addressing management of locally manufactured, developed or modified tools and other support equipment. **(T-3).** All locally manufactured, developed or modified tools and equipment must be approved by the commander/designated representative. **(T-3).** Examples of these items are test fixtures. This procedure does not apply to local manufacture, modification or design of tools authorized in specific technical data. **(T-3).**

13.4.3. The work center will establish procedures for tracking and using UTC/mobility tool kits as follows:

13.4.3.1. Inspect/inventory and document master tool inventory on UTC/mobility tool kits/tools when opened, prior to resealing for storage, or at a minimum, quarterly in

garrison. **(T-3).** If resources do not permit designated UTC/mobility tool kits, garrison kits can be used.

13.4.3.2. Inventory and seal the UTC/mobility tool kits to eliminate use while in garrison. **(T-3).**

13.4.3.3. The seal on the UTC/mobility tool kits will show date it was sealed and who completed inventory. **(T-3).**

13.4.3.4. Mark the UTC/mobility toolboxes in accordance with AFI 10-403 (e.g., packing list). **(T-3).**

13.4.4. When addressing the management of industrial shop machinery tools and accessories/attachments (e.g., Hiltis, band saws, arbors, chucks, portable vise, blades, Pionjars) that normally do not fit within the tool kit the work center will:

13.4.4.1. Develop master tool inventory reflecting all industrial shop machinery tools and accessories/attachments that are not stored within the tool's container. **(T-3).** Inventory includes storage location of tool.

13.4.4.2. If a tool has multiple pieces stored within its container, create an inventory of the contents (e.g., Hilti's drill box contains one drill, two 1/2-inch and one 3/4-inch 6-inch bits, and 3 ounces of lubricating oil). **(T-3).**

13.4.5. Consider the following when establishing a tool inventory tracking system that facilitates quick inventory and accountability of tools:

13.4.5.1. Design tool kits to provide a quick inventory and accountability of tools.

13.4.5.2. Develop a simple inventory method, a "show" (e.g., a shadow of the tool) and "know" (knowledge of tool or kit location) concept.

13.4.5.3. Establish tool kit contents for ease of inventory. Tool kit contents should be standardized to the maximum extent possible within functional areas within a unit that have similar missions.

13.4.5.4. Each tool, equipment item, or consumable contained in a tool kit that can have an assigned location identified will have one of the following: inlay cuts in the shape of the item, shadowed layout, label, silhouette or drawer/location assignment. **(T-3).** No more than one item will be stored in a cutout, shadow, or silhouette except for tools issued in sets such as drill bits, Allen wrenches, apexes or paired items (e.g., gloves, booties). **(T-3).**

13.4.5.5. Utility tool kits (e.g., green tool bags, briefcase kits) do not have assigned/marked individual tool locations within the utility tool kit; however, the master inventory must reflect where the utility tool kits are stored when not in use. **(T-3).**

Chapter 14

CORROSION PREVENTION AND CONTROL PROGRAM (CPCP)

14.1. Introduction.

14.1.1. Corrosion is a natural phenomenon that attacks metal by electrochemical action and converts the metal into a metallic compound, such as an oxide, hydroxide, or sulfate. Corrosion occurs because of the tendency for metals to return to their natural state.

14.1.2. Corrosion, if left unchecked, progressively degrades an item's strength until its structure can no longer sustain its design load and/or cause malfunctions. Unchecked corrosion can result in excessive maintenance and repair as well as system downtime and product contamination.

14.2. Maintenance Work Center Functions.

14.2.1. Effective and timely equipment corrosion prevention and control actions will be performed in accordance with TO 31Z-10-37, *Corrosion Prevention and Protection*, applicable TM/TO, and other command instructions. **(T-3).**

14.2.2. All work center technicians will complete AFJQS XXXX-201C, *Corrosion Prevention and Control*, and document in training records in accordance with AFI 36-2670. **(T-3).** If local tasks are required, tasks will be identified, added to the training plan, and training conducted. **(T-3).**

14.2.3. Submissions that address the type, extent, and repair/treatment of corrosion on equipment will be provided through accurate and timely maintenance and historical documentation. **(T-3).**

14.2.4. If deemed necessary, research, develop, and publish corrosion-control procedures tailored to work center unique needs or high corrosion prone areas. **Note:** It is important to identify corrosion-prone areas, such as dissimilar metals, fasteners, hinges, latches, rivets, mating surfaces, crevices, faying surface, fillet sealing, spot-wielding assemblies, and water entrapment areas.

14.2.5. Equipment corrosion-control PMI intervals shall be modified/changed to meet the corrosion environment in which the equipment will be operating. **(T-3).** Some locations may need to clean equipment more frequently.

14.2.6. LWCs and PMIs will be verified for equipment containing clear corrosion-control inspection requirements and procedures. **(T-3).** Consider pertinent factors including operational environment, cost, resources, availability, and probability to establish inspection interval.

14.2.7. Guidance and resolution of corrosion problems beyond the unit's capability to repair will be coordinated with parent MFM/ MAJCOM FAM. **(T-3).**

Chapter 15

HISTORICAL RECORD MANAGEMENT

15.1. Introduction.

15.1.1. This chapter prescribes general requirements and procedures for the administration of maintenance documents, equipment inspections, equipment transfers, and equipment historical data. Work center facility, systems installation, and equipment records are historical documentation that constitute a permanent record of significant maintenance actions or significant changes to the facility or system.

15.1.2. Historical records must remain with each equipment or system throughout its life cycle. **(T-3).** They are designed to provide technicians and Item Managers (IM) with an accurate portrayal of significant actions, modifications, issues, and overhauls. System/equipment historical files will be maintained in the work center in a centralized file in accordance with AFI 33-322, or other applicable directives. **(T-3).**

15.1.3. Records will be disposed in accordance with AFRIMS RDS, or other applicable directives. **(T-3).** Electronic files are authorized. For classified reporting, the approved Air Force /MAJCOM/FOA/DRU security classification guide will be followed for applicable equipment/system. **(T-3).**

15.2. Overview.

15.2.1. Units will develop master inventories based on shipping, acceptance, catalog, and other applicable documents for Air Force, MAJCOM, FOA, DRU, or locally procured COTS/GOTS equipment if an AFTO Form 470 has not been established per TO 00-35D-2. **(T-3).**

15.2.2. Historical records, AFTO Form 95, *Significant Historical Data*, IMDS history files, or other records shall be required on all MICAP-reportable equipment, command-supported equipment, and other equipment as designated by the IM/SM/PM. **(T-3).** IMs shall determine which items in the Mission Design Series (MDS)-specific -6 manual require historical reporting and inform the SM so the equipment/system can be identified for reporting. **(T-3).**

15.2.3. For ALC-managed systems/equipment, AFTO Forms 95 or IMDS history files shall be required on all SRD-coded systems/equipment except for Communications Security equipment, command-supported equipment, and other equipment as designated by the IM/SM/PM. **(T-3).** The IMs shall determine which equipment items specific -6 manual historical reporting is required and inform the SM so the equipment/systems can be identified for reporting. **(T-3).**

15.2.4. For non-ALC-managed systems/equipment, historical records or equivalent information will be maintained utilizing AFTO Form 95. **(T-3).** Each item does not require its own record; technicians may establish records based on location. Records may be maintained in either electronic or manual forms.

15.2.5. All equipment/systems tracked in IMDS must utilize the IMDS history files and record all significant maintenance actions; especially TCI, TCTOs, and Time Compliance Network Orders. **(T-3).** In addition, all tactical and mobile equipment will require historical records.

(T-3). The records or record duplicates must accompany the equipment when deployed. **(T-3).**

15.2.6. Historical records must accompany equipment sent to depot for repair/overhaul and transferred between units. **(T-3).**

15.2.7. For more information, refer to TO 00-20-1. These chapters apply to all Air Force personnel, equipment/systems, and contractors.

15.3. Work Center Functions. Work centers ensure facility, systems installation, and equipment records are current and available. Work centers will:

15.3.1. Use IMDS to collect historical data. **(T-3).** If IMDS is not available, manually collect the information with enough detail to satisfy IMDS reporting requirements when the system is back up. **(T-3).**

15.3.2. Use AFTO Form 95 or its automated equivalent. **(T-3).**

15.3.3. Create historical data upon the issuance of the first TCTO or the occurrence of the first condition or incident requiring data entries. **(T-3).**

15.3.4. Maintain permanent history and historical files. **(T-3).**

15.3.4.1. Maintain a permanent history on end items listed on the SRD table in IMDS, command-supported equipment designated by the MAJCOM, FOA, or DRU; and other critical systems assigned to the organization that are not tracked in IMDS. **(T-3).**

15.3.4.2. Maintain system/equipment historical files in the work center in a centralized file in accordance with AFI 33-322, and dispose of in accordance with AFRIMS RDS. **(T-3).** Electronic files are authorized. **Note:** RMC Supported equipment historical file will be maintained by RMC personnel. **(T-3).**

15.3.4.3. Establish and maintain an individual historical file in accordance with AFI 33-322, for each designated equipment end item per applicable Air Force directive. **(T-3).**

15.3.4.4. Centrally locate historical document files in the documentation activity of the unit possessing/maintaining the equipment (e.g., one file cabinet). **(T-3).** Online computer systems are considered centrally located files and authorized. **(T-3).**

15.3.4.5. Include hard copy or electronic historical document files for subsystems and components in equipment end item files or maintain them in a separate file. **(T-3).**

15.3.4.6. Consolidate files for noncomplex items into a single folder or a series of folders. **(T-3).** Each individual file will contain historical documents, operational data, maintenance status documents, and reports that reflect current status. **(T-3).**

15.3.5. Document BCE facility grounding and lightning protection checks in the facility record or historical files. **(T-3).** Work centers will submit an AF Form 332, *Base Civil Engineer Work Request,* to BCE every 21 months per AFI 32-1065, Table 1, to ensure ground check frequency is maintained. **(T-3).** Work centers will schedule and complete grounding and lightning protection for facilities that are maintained by the RMCs. **(T-3). Note:** Work centers will forward grounding and lightning protection check documents to the RMC for RMC supported equipment. **(T-3).** RMC will add documents to facility record or historical files. **(T-3).**

15.3.6. Transfer documents with all equipment being transferred to another organization. **(T-3).**

15.3.6.1. Verify all current maintenance and historical documents or computer-generated equivalents accompany the equipment or are forwarded to the new maintenance activity electronically, no later than the same day the transfer occurs. **(T-3).** Records may be burned onto a Compact Disk-Read Only Memory (CD-ROM) or Digital Versatile Disc-Read Only Memory (DVD-ROM).

15.3.6.2. Ensure applicable documents are placed in a waterproof envelope and securely attached to the component, item or container, when end items that require separate historical files are transferred as separate units. **(T-3).** If the item is not packaged or crated, the waterproof envelope will be securely attached to the item in a location that will provide the best protection from exposure to the elements and prevent loss during handling and/or mail a copy to the gaining organization to prevent loss. **(T-3).**

15.3.7. Use and maintain TCTO information. **(T-3).**

15.3.7.1. Print a copy of the Automated History Entry (AHE) record or provide an electronic copy in the applicable e-file in units using IMDS AHE. **(T-3).** Ensure the history, whether an electronic version on a CD-ROM or paper copy, accompanies the equipment/system being turned in or transferred to another unit or going to depot. **(T-3). Note:** Do not delete the AHE in IMDS if the equipment is being transferred. Instead, transfer the records to the gaining organization in accordance with AFCSM 21-567, Volume 2, *Equipment/Personnel Transfer Software User Manual.* **(T-3).**

15.3.7.2. Verify equipment/systems gained and loaded into IMDS have the AHE block annotated. **(T-3).** See AFCSM 21-568, Volume 2, for details. This allows the TCTO history to be recorded in the AHE.

15.3.7.3. Print a copy of the TCTO history from the IMDS TCTO subsystem and attach it to the AFTO Form 95 in units using manual AFTO Form 95 instead of AHE. **(T-3).**

15.3.8. Print or type all entries on historical documents, with the exception of signatures (when available, electronic signatures are authorized). **(T-3).** If electronic signatures are not available in IMDS or approved form and the record is maintained electronically, the first name, last name, rank, organization/office symbol will be included. **(T-3).**

15.3.9. Immediately notify the shipping/losing organization when equipment is received, and the historical documents are missing or contain incomplete information. **(T-3).**

15.3.9.1. The shipping/losing organization will promptly forward the missing documents or provide all available information for completion of the documents for initiation of new documents. **(T-3).**

15.3.9.2. Contact the applicable SM through the MAJCOM, FOA, or DRU for instructions when the documents cannot be located, and the asset serviceability cannot be readily assessed. **(T-3).**

15.3.10. Maintain historical documents on equipment in extended storage. **(T-3).**

15.3.10.1. Historical documents will be filed with the equipment or in the centralized file if the equipment is stored at the unit/base level. **(T-3).**

15.3.10.2. Prepare up-to-date maintenance and historical documentation for each piece of equipment being returned to service. **(T-3).**

15.3.10.3. When equipment is maintained in extended storage at an organization or activity, the commander may request a waiver through the respective MAJCOM, FOA, or DRU on a case-by-case basis. All approved waivers must be maintained in the equipment's historical documents. **(T-3).**

15.3.11. Record all dates on the required forms using 8 digits in the order of year, month, and day. Example: YYYYMMDD, 20160418 for 18 Apr 2016. **(T-3). Note:** Approved automated forms in IMDS may deviate from this procedure based upon system standards. Handwritten formats may follow the IMDS format.

15.3.12. Only use electronic signatures to sign off maintenance documentation where the records are protected with password and ID and authorization is limited to those individuals as outlined in the 00-20-series TMs or other directives. **(T-3).**

15.3.13. Ensure the processing of documents during depot maintenance follow the correct procedures. **(T-3).** Documents will be processed in accordance with TO 00-25-108 and the 00-20-series TMs. **(T-3).** Personnel preparing organic or contract work statements will ensure that the provisions of these TMs are included in all applicable maintenance contracts. **(T-3).**

15.3.14. Maintain records on mobile equipment. **(T-3).**

15.3.14.1. Maintain one AFTO Form 95 or automated equivalent on each mobile facility to record significant historical data; however, in cases where a facility is comprised of more than one van, an AFTO Form 95 or automated equivalent will also be maintained on each van. **(T-3).**

15.3.14.2. Maintain an AFTO Form 95 or automated equivalent on each end item of equipment not permanently assigned to a van or facility. **(T-3). Note:** These items have their own SRD and are listed with a WUC under the van or facility WUC.

15.3.14.2.1. Ensure all entries are transcribed to the AFTO Form 95 for the assigned van or facility, if a non-permanent item is permanently transferred to a van or facility. **(T-3).**

15.3.14.2.2. Initiate a new document if a permanent item removed from a van or facility becomes its own end item. **(T-3).** Document transfer of pertinent information to the new record and annotate in the original record the removal reason and date. **(T-3).**

15.3.14.2.3. A duplicate copy of historical documents may accompany all deployed equipment and appropriate entries will be accomplished on the documents while deployed, if the original historical documents do not accompany the equipment when deployed. **(T-3).** Use of a duplicate record is at the unit commander/flight commander's discretion.

15.3.14.2.3.1. Ensure the original documents will be updated to reflect events from the deployment if duplicate records are used when the equipment is returned to the home station. **(T-3).**

15.3.14.2.3.2. Ensure both files match upon deployment if duplicate records are

used. **(T-3).**

15.3.15. Ensure applicable AFTO Form 470-series checklists (if available), automated copies, or master COTS inventories accompany equipment items being turned in to supply, transferred to another agency, or deployed. **(T-3).** Refer to TO 00-35D-2 for additional guidance on use of AFTO Form 470-series.

15.3.16. Units will develop master inventories based on shipping, acceptance, catalog, and other applicable documents for COTS/GOTS equipment. **(T-3).** When establishing the master COTS inventories, include the initial quantity received, date received, serial number if applicable, part number, kit number/box number, and description. **(T-3).**

15.3.17. Ship historical documents with equipment or component to disposal, storage activity, next using activity or depot unless otherwise directed. **(T-3).**

15.3.18. Maintain current drawings. **(T-3).**

15.3.18.1. Each maintenance work center must document and maintain outside plant drawings for its area of responsibility in accordance with TO 00-33D-3004. **(T-3).**

15.3.18.2. Ensure all facility drawings are placed into the applicable AIS and verify an updated copy of facility record drawings needed for system troubleshooting is available in the work center or in an easily accessible central location within the complex. **(T-3).**

15.4. Completion of Air Force Technical Order Form 95.

15.4.1. The AFTO Form 95 is a permanent document of significant actions and provides the maintenance organization with a life profile of the item. This information portrays conditions that could have a bearing on future maintenance of the equipment.

15.4.2. At a minimum, installation/removal dates, component accumulated hours, reason for removal, and a brief narrative as to the maintenance performed on the component (e.g., unit overhauled; unit cleaned; inspected and repaired; replaced minor parts; TCTOs completed; and scheduled maintenance complied with) will be annotated. **(T-3).**

15.4.2.1. Block 1, MDS or type designator of the weapon system or equipment. Enter the part number assigned to the item.

15.4.2.2. Block 2, “Manufacturer.” Enter the name of the manufacturer.

15.4.2.3. Block 3, “Serial Number.” When assigned, enter the serial number of the item identified in block 1. Example: 85-1428, 64-14828. **Note:** For systems that have multiple pieces/boxes and no data plate identifying the serial number, prior to establishing the equipment record and associated historical, verify how to identify the item in IMDS with the Item Manager or Lead Command.

15.4.2.4. Block 4, “Acceptance Date.” Enter the date the equipment was accepted by the Air Force. If unknown, enter “unknown.” Generally, this is the date the first organization to get the equipment accepts it.

15.4.2.5. Column A, “Date.” Enter the date the significant event such as maintenance action or inspection occurred.

15.4.2.6. Column B, “Remarks”. Enter the applicable information, using as many lines as necessary, to document significant data.

15.5. Historical Records. At a minimum, historical records will contain the following:

15.5.1. Applicable AFTO 470 or equivalent. **(T-3).**

15.5.2. TCTO Compliance, if not recorded in IMDS. **(T-3).** Document a TCTO noncompliance due to modified or removed systems in which an AF Form 1067 was approved by MAJCOM, FOA, or DRU. **(T-3).** All AF Form 1067 must be maintained in the historical files. **(T-3).** Annotate an entry stating a TCTO was received and whether or not it is applicable. **(T-3).** If the TCTO is deemed applicable, annotate compliance and when the kits were ordered, if necessary. **(T-3).**

15.5.3. TCIs when not recorded in IMDS. **(T-3).**

15.5.4. Removal and replacement of critical subsystems or items or WUC items to include the reason for removal and a brief narrative as to the maintenance performed on the component (e.g., unit overhauled, unit cleaned, inspected and repaired, corrosion, replaced minor parts, TCTOs completed, DIFM, exchange of end item [controlled substitution], scheduled maintenance complied with). **(T-3).**

15.5.5. Remarks concerning special service test equipment installed or removed. **(T-3).**

15.5.6. Information regarding corrosion: location, extent, and treatment accomplished or required. **(T-3).**

15.5.7. Circumstances regarding mishaps: extent of damage, repair authority, repairing activity, date of repair, and repairs accomplished. **(T-3).**

15.5.8. Weather damage to equipment. **(T-3).**

15.5.9. Data on chemical, biological or radiological contamination: contamination date, type of contaminant, decontamination date, type of decontaminants, and decontamination procedures used. **(T-3).** Records will be maintained for the lifecycle of the equipment (including removed/installed parts). **(T-3).**

15.5.10. Any variance records (e.g., AF Form 1067), approved and not approved, that change the standard configuration in accordance with TM/commercial manual or original installation paperwork. **(T-3).**

15.5.11. If the facility or equipment requires a flight check before commissioning, enter the initial flight check data, together with any significant maintenance actions that were required to pass initial flight check. **(T-3).**

15.5.12. The removal or decommissioning of end items of equipment and the date the equipment was packed for shipment or the date the equipment was placed in storage as applicable. **(T-3).**

15.5.13. The removal and replacement of items resulting from excessive contaminants discovered through the Spectrometric Oil Analysis Program. **(T-3).**

15.5.14. Remarks concerning special service tests and special test equipment installation and removal are required. **(T-3).**

15.5.15. All circumstances regarding accidents or incidents. **(T-3).** The extent of damage and repairs accomplished. **(T-3).**

15.5.16. Significant maintenance action and circumstances involving emergency maintenance accomplished by depot, engineering installation squadrons, special communications teams, or contractor maintenance personnel. **(T-3).**

15.5.17. TCTO actions, when not under the mechanized system (TO 00-5-15, Section II). **(T-3).**

15.5.18. Date and signature showing that the annual review of historical records was conducted and that they reflect all required actions and are accurate and current. **(T-3).**

15.5.19. Document chronic problems and any condition or maintenance action that could affect future maintenance of the equipment. **(T-3).**

15.5.20. Copies of approved system and site-specific waivers and deviations. **(T-3).**

Chapter 16

CONTRACT MAINTENANCE

16.1. Introduction. This section identifies the basic responsibilities for managing Most Efficient Organization or contract maintenance.

16.2. Compliance. Laws and applicable regulations will be complied with. **(T-0)**. Executive orders, instructions, and TOs will be executed by the government unless waived by the appropriate authority. **(T-3).** When the government solicits services via contract, some requirements may be excluded from the Performance-Based Work Statement (PWS) or Statement of Work (SOW) due to cost. Requirements excluded from the PWS or SOW will be performed by the government. **(T-3).** MFM's will determine which functions that are not included in the PWS or SOW will be waived or completed by other means. **(T-3).**

16.3. Contract Maintenance Management Responsibilities. When maintenance management responsibilities are required of a contractor, they must be specifically identified in the PWS. **(T-3).** Managers will base PWS requirements on existing maintenance policy. **(T-3).** The intent and philosophy of organic maintenance processes must be carefully considered and included when appropriate. **(T-3).** Contract maintenance personnel are required to document maintenance actions and order parts, the preferred method is through the Air Force-approved AIS, to include IMDS and Reliability and Maintainability Information System (REMIS). **(T-3).** A contractor may generate their own database to track these actions. As a minimum, PWSs shall have:

16.3.1. Contractor Maintenance Data Collection. **(T-3).** Enter systems and component failure in accordance with TO 00-20-2; material consumption in accordance with TO 00-20-3, and TCTO reporting requirements in accordance with TO 00-5-15. **(T-3).** Cite appropriate data items and include collecting TCTO and reparable processing data in contracts. **(T-3).** Contract instructions call out the time and place to turn in data. **(T-3).** Specify if the contractors provide complete source documents and forms or automated products. **(T-3).**

16.3.2. System restoral priorities. **(T-3).**

16.3.3. Procedures to maintain configuration control. **(T-3).**

16.3.4. Procedures to document historical records, to include procedures if automated documentation method is unavailable. **(T-3).**

16.4. Contract Surveillance Programs/Contracting Office Representative. Contract Surveillance Programs/Contracting Office Representative accomplish contract maintenance surveillance program duties in accordance with procurement guidelines and command directives. **(T-3).** When requested by the procurement office, the commander/superintendent provides a qualified Contracting Office Representative for surveillance purposes. **(T-3).** Surveillance programs may vary depending on the scope of the contract and directions of the procurement office. It is vital military personnel do not change or task contractors thus changing the contract. **(T-3).**

16.5. Contracting Officers. Only contracting officers shall enter into or change a contract, PWS or SOW. **(T-0).** Only contracting officers or Contracting Office Representative s may provide direction to contractors. **(T-0).** Direct problems with contract performance, contract changes or interpretation to the responsible contracting officer, administrative contracting officer or

Contracting Office Representative for resolution. **(T-0).** Contractual assistance may be used to satisfy a wide variety of requirements ranging from minor one-time repairs to operation and maintenance of complete systems. For more information see AFI 63-138, *Acquisition of Services,* and AFPD 64-1, *The Contracting System.*

16.6. Considerations. Consider contract maintenance on a planned and selective basis; however, wartime capability shall not be compromised by excessive use of this resource. **(T-3).** Check with appropriate MAJCOM, FOA, or DRU managers to ensure local contract efforts do not erode command wartime capabilities.

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(AETC)

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Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

AFCSM 21-561, Volume 2, *Maintenance Events Software User Manual,* 1 February 2019

AFCSM 21-567, Volume 2, *Equipment/Personnel Transfer Software User Manual,* 1 February 2019

AFCSM 21-568 Volume 2, *Time Compliance Technical Order (TCTO) Software User Manual,* 1 June 2019

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AFI 10-601, *Operational Capability Requirements Development,* 6 November 2013

AFI 13-201, *Airspace Management,* 21 August 2012

AFI 13-217, *Drop Zone (DZ) and Landing Zone (LZ) Operations*, 9 May 2007

AFI 21-103, *Equipment Inventory, Status and Utilization Reporting,* 20 January 2020

AFI 23-101, *Air Force Materiel Management,* 11 December 2016

AFI 23-111, *Management of Government Property in Possession of the Air Force,* 18 November 2018

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None

(Added-AETC) None

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AF Form 1067, *Modification Proposal*

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AF Form 2419, *Routing and Review of Quality Control Reports*

AF IMT 2420, *Quality Control Inspection Summary*

AFTO Form 22, *Technical Manual (TM) Change Recommendation and Reply*

AFTO Form 95, *Significant Historical Data*

AFTO Form 217, *Certificate of Maintenance Accomplished*

AFTO Form 227, *C-E Depot Maintenance Requirements and Schedule*

AFTO Form 229, *Engineering Installation Assistance Request*

AFTO Form 349, *Maintenance Data Collection Record*

AFTO Form 470, *Electronic Set Inventory Checklist*

AFTO Form 471, *Electronic Set Inventory Checklist Configuration Data*

AFTO Form 472, *Electronic Set Inventory Checklist Completion Data*

AFTO Form 747, *Cyberspace Infrastructure System Acceptance*

DD Form 1475*, Basic Allowance for Subsistence-Certification*

FAA Form 8240-22, *Facility Data*

Abbreviations and Acronyms

AETC—Air Education and Training Command

AF—Air Force

AFCFM—Air Force Career Field Manager

AFCSM—Air Force Computer Systems Manual

AFEMS—Air Force Equipment Management System

AFETS—Air Force Engineering and Technical Services

AFFSA—Air Force Flight Standards Agency

AFH—Air Force Handbook

AFI—Air Force Instruction

AFIS—Air Force Inspection System

AFJQS—Air Force Job Qualification Standard

AFMAN—Air Force Manual

AFMC—Air Force Materiel Command

AFMCI—Air Force Materiel Command Instruction

AFMQCC—Air Force Maintenance Quality Control Checklist

AFMSI—Air Force Maintenance Special Instructions

AFPAM—Air Force Pamphlet

AFPD—Air Force Policy Directive

AFQTP—Air Force Qualification Training Package

AFREP—Air Force representative

AFRIMS—Air Force Records Information Management System

AFSC—Air Force Specialty Code

AFTO—Air Force Technical Order

AHE—Automated History Entry

AI—Acceptance Inspections

AIS—Automated Information System

ALC—Air Logistics Center

ANG—Air National Guard

AOR—Area of Responsibility

ART—Air and Space Expeditionary Force Reporting Tool

AS—Allowance Standards

ATC—Traffic Control

ATCALS—Air Traffic Control and Landing Systems

AWP—Awaiting Parts

BAS—Basic Allowance for Subsistence

BCE—Base Civil Engineer

C—Complies

C-E—Communications Electronics

CA/CRL—Custodian Authorization Custody Receipt Listing

CAFDEx—Centralized Access for Data Exchange

CDC—Career Development Course

CDE—Corrected During Evaluation

CD-ROM—Compact Disk-Read Only Memory

CES—Civil Engineering Squadron

CFETP—Career Field Education and Training Plan

CIPS—Cyberspace Infrastructure Planning System

COM—Chief of Maintenance

CONUS—Continental United States

COR—Critical Outage Report

COTS—Commercial Off-The-Shelf

CPCP—Corrosion Prevention and Control Program

CWC—Complies With Comments

DASR—Digital Air Surveillance Radar

DATCALS—Deployable Air Traffic Control and Landing Systems

DIFM—Due-In From Maintenance

DoD—Department of Defense

DPEM—Depot Purchased Equipment Maintenance

DRU—Direct Reporting Unit

EAO—Equipment Accountability Officer

ECO—Equipment Control Officer

EE—Equipment Evaluation

E&I—Engineering and Installation

EMWG—Encroachment Management Working Group

EPE—Equipment Personnel Evaluation

ESD—Electrostatic Discharge

ESR—Equipment Status Reporting

ETIMS—Enhanced Technical Information Management System

E—Tools—Electronic Tools

ETRO—Estimated time of return to operation

FAA—Federal Aviation Administration

FAM—Functional Area Manager

FMC—Fully Mission Capable

FOA—Field Operating Agencies

GOTS—Government Off-The-Shelf

HHQ—Higher Headquarters

HQ—Headquarters

I&E—Inspections and Evaluations

ID—Identification

IG—Inspector General

IM—Item Manager

IMDS—Integrated Maintenance Data System

IPE—Individual Personnel Evaluation

IT—Information Technology

ITP—Individual Training Plan

JCN—Job Control Number

JDD—Job Data Documentation

JO—Joint Orders

LCMP—Life Cycle Management Plans

LCSE—Life Cycle Systems Engineering

LMQCC—Local Maintenance Quality Control Checklist

LOTO—Lock Out Tag Out

LRS—Logistics Readiness Squadron

LWC—Local Work Card

MAJCOM—Major Command

MARR—Maintenance Action Review Report

MDS—Mission Design Series

MFM—MAJCOM Functional Manager

MICAP—Mission Capable

MICT—Management Internal Control Toolset

MISCAP—Mission Capability

MRT—Mitigation Response Team

MOA—Memorandum of Agreement

MOC—Maintenance Operations Center

MQCC—Maintenance Quality Control Checklist

MTP—Master Training Plan

NAVAID—Navigational Aids

NCOIC—Noncommissioned Officer In Charge

NEXRAD—Next Generation Radar

NFPA—National Fire Protection Association

NIC—Not In Compliance

NMC—Not Mission Capable

NOTAM—Notice to Airmen

NRTS—Not Repairable This Station

OCR—Office of Coordinating Responsibility

OCCR—Organizational Cost Center Record

OE/AAA—Obstruction Evaluation/Airport Airspace Analysis

OI—Operating Instruction

OIL—Open Incident Listing

OJT—On-the-Job Training

OPR—Office of Primary Responsibility

OSS&E—Operational Safety, Suitability, and Effectiveness

PCW—Previously Completed With

PE—Personnel Evaluations

PFMR—Project Fund Management Record

PM—Program Manager

PMEL—Precision Measurement Equipment Laboratory

PMI—Preventive Maintenance Inspection

POC—Point of Contact

PPLAN—Programming Plan

PWS—Performance-based Work Statement

QA—Quality Assurance

QAR—Quality Assurance Representatives

RAWS—Radar, Airfield, and Weather Systems

RDS—Records Disposition Schedule

RMC—Regional Maintenance Center

RMWC—Regional Maintenance Work Center

ROC—Radar Operations Center

RSEP—Requirements Scheduling and Execution Program

RSP—Readiness Spares Package

SAA—Senior Airfield Authority

SAPM—Self-Inspection/Assessment Program Manager

SDR—System Documentation Releases

SM—System Manager

SME—Subject Matter Expert

SORTS—Status of Reporting and Training System

SOW—Statement of Work

SPE—Special Personnel Evaluation

SPECAT—Special Duty Catalog

SRD—Standard Reporting Designator

SSM—System Support Modifications

STARS—Standard Terminal Automation Replacement System

TBA—Training Business Area

TCI—Time Change Item

TCTO—Time Compliance Technical Order

TI—Technical Instructions

TM—Technical Manual

TMDE—Test, Measurement and Diagnostic Equipment

TO—Technical Order

TODA—Technical Order Distribution Account

TODO—Technical Order Distribution Office

TRN—Turn-Around

UEI—Unit Effectiveness Inspection

UND—Urgency of Need Designator

UTC—Unit Type Code

UTM—Unit Training Manager

WSR-88D—Weather Surveillance Radar-1988, Doppler

WUC—Work Unit Code

Terms

Aerodrome—A defined area on land or water (including any buildings, installations and equipment) intended to be used either wholly or in part for the arrival, departure, and movement of aircraft.

Air Traffic Control and Landing Systems—DoD facilities, personnel, and equipment (fixed, mobile, and seaborne) with associated avionics to provide safe, orderly, and expeditious aerospace vehicle movements worldwide.

Career Field Education and Training Plan—The CFETP is a comprehensive core- training document that identifies: life-cycle education and training requirements; training support resources, and minimum core task requirements for a specialty. The CFETP aims to give personnel a clear path and instill a sense of industry in career field training. It is the formal training contract between the Air Force Career Field Manager and AETC for formal accession and life- cycle skills training.

Flight line—Any area or facility including apron, hardstand and ramps on or in which aircraft may be parked, stored, serviced or maintained.

Ground Radar and Airfield Equipment/Systems—Defined as ATC, weather, ground aircraft control and warning Radar systems; related Radar operator training devices; aircraft identification equipment; remoting systems; video mappers; computerized processors; meteorological; navigational aids; and ATC radio and communications systems and facilities, fixed and deployable. In this document, "Ground Radar and Airfield systems" are referred to as equipment/system. This type of equipment/system is normally tracked in an Air Force approved formal accounting/tracking system such as the IMDS, AFEMS CA/CRL or the AFEMS Asset Inventory Management system. This equipment generally has defined preventive maintenance inspections and other sustainment programs in place.

Installation Commander—The individual responsible for all operations performed by an installation.

Major Command (MAJCOM)—For the purpose of this manual, includes all Air Force Major Commands plus the Air National Guard Readiness Center, Air Force Reserve Command, Direct Reporting Units, and Field Operating Agencies.

Master Training Plan—Employs a strategy for ensuring the completion of all work center job requirements by using a Master Task Listing and provides milestones for task, CDC completion, and prioritizes deployment/UTC, home station training tasks, upgrade, and qualification tasks.

Mishap—A mishap is an unplanned occurrence, or series of occurrences, that results in damage or injury and meets Class A, B, C, D and Class E event reporting criteria in accordance with AFI 91-204, *Safety Investigations and Reports,*paragraph 1.10. Damage or injury includes: damage to DoD property (excluding normal wear and tear or aging); occupational illness to DoD military or civilian personnel; injury to DoD military personnel on- or off-duty; injury to on-duty DoD civilian personnel; damage to public or private property, or injury or illness to non-DoD personnel caused by Air Force operations.

Readiness Spares Package—Items other than aircraft, such as C-E, vehicles, and bare base systems, that are a transportable set of spares and repair parts required to support planned wartime or contingency operations for a specified period of time pending resupply.

Objective—A statement that specifies what behavior is to be exhibited, the conditions under which behavior will be accomplished and the minimum standard of performance. Objectives describe only the behaviors that directly lead to or specifically satisfy a job performance requirement. An objective is a statement of instructional intent.

On—the-Job Training—Hands-on, over-the-shoulder training conducted to certify personnel in both upgrade and job qualification training.

Qualified Personnel—Refers to those individuals with the technical skills required to perform various functions relating to Ground Radar and Airfield equipment/system corrective, inspection, maintenance, and preventive maintenance activities. An individual is considered "qualified" if that individual has been trained to the level necessary to perform specific activities or functions under this TO. Technical qualifications will vary by function being performed and technical competency required. For example, an individual qualified to troubleshoot a system will require more specialty/technical training than an individual qualified to remove and install a part. An individual qualified to remove and install a part will require more specialty/technical training than a custodian who conducts operational inspections and assessments.

Qualification Training Package—An instructional package designed for use at the unit to qualify, or aid qualification, in a duty position or program, or a piece of equipment. It may be printed, computer-based, or in other audiovisual media. AFQTPs do not require third-party.

Skill Level—The level of qualification within an awarded Air Force specialty, shown by the fourth digit of the AFSC.

Specialty Training Standard—An Air Force publication that describes an Air Force specialty in terms of tasks and knowledge which an airman in that specialty may be expected to perform and identifies the training provided to achieve a 3-, 5-, or 7-skill level within an enlisted Air Force. It further serves as a contract between AETC and the functional user to show the overall training requirements for an AFSC that are taught in formal schools and correspondence schools.

Special Duty Catalog—The purpose of the SPECAT Guide is to provide additional eligibility criteria for specific duties and eligibility requirements for assignments filled through the Developmental Special Duty (DSD) process and EQUAL-Plus. The SPECAT is applicable to RegAF only. It is not applicable to ANG or Air Force Reserve Airmen.

Supplemental Training—Training for a portion of an Air Force Specialty without a change in AFSC. Formal training on new equipment, methods and technology that is not suited for on-the-job training.

Task—A unit of work activity or operation that forms a significant part of a duty. A task usually has clear beginning and ending points and directly observable or otherwise measurable processes, frequently but not always resulting in a product that can be evaluated for quantity, quality or fitness in the work environment. A task is performed for its own sake; that is, it is not dependent upon other tasks, although it may fall in a sequence with other tasks in a duty or job array.

Attachment 2

EVALUATION REPORT WRITING AND SUBMISSION

A2.1. Personnel Evaluation Report Writing.

A2.1.1. To determine overall results, evaluators will review notes and captured examples, then validate references. **(T-3).** Results are based on overall process performance.

A2.1.1.1. Evaluators must document all evaluation task deficiencies including Corrected During Evaluation (CDE) corrections. **(T-3).**

A2.1.1.2. Evaluators will use standardized codes, to include CDEs, to track all deficiencies for trend analysis. **(T-3).** The decision to document a deficiency must be based on published standard practices and procedures, technical data, and other policy guidance. **(T-3).**

A2.1.1.3. Evaluators will include a deficiency code and a category rating of either critical (Cat I), major (Cat II); or minor (Cat III) based on **Table A2.1** for all validated deficiencies. **(T-3).**

A2.1.1.4. Evaluators will rate each task included in the overall evaluation. **(T-3).** All deficiencies are then totaled and assigned an overall task result in accordance with **Table A2.2,** Personnel Evaluation Task Result Matrix. **(T-3).** Results are based on the following three-grade system; C; CWC, or NIC, per **Table A2.2** Technicians will be decertified on task(s) rated NIC. **(T-3).**

A2.1.1.5. **Table A2.1.** shows what deficiency code may apply to a certain situation.

A2.1.1.6. **Table A2.2.** shows how to determine the results based on the deficiency code results for each task evaluated.

A2.1.1.6.1. If a technician sets up an oscilloscope that is overdue calibration and does not use it, document this mistake as a Cat III.

A2.1.1.6.2. If the technician sets up an oscilloscope that is overdue calibration and uses the oscilloscope during the evaluation to make measurements in accordance with specifications outlined within a T.O. or technical guidance, then the evaluator documents the mistake as a Cat I or II deficiency. Determination of Cat I or II is dependent on how serious the impact might be on system or equipment capability by using potentially out-of-tolerance test equipment. Based on the evaluator's assessment using **Table A2.1** and **Table A2.2**, the task result would be either a CWC or a NIC for that task. A NIC would require the member be decertified on the task and reevaluated within 90 calendar days. A CWC result allows the supervisor based on the evaluator recommendation to determine if the member requires task decertification. **Note: Table A2.1** provides an objective method to evaluate errors. Nevertheless, the evaluator's experience and subjective judgment plays a part in the assessment.

A2.1.1.7. PEs that contain no (zero) NIC tasks are rated as "Satisfactory." Evaluations that contain one or more NIC tasks are rated as "Unsatisfactory."

A2.1.1.8. Brief the technician and the work center supervisor upon determining the evaluation resulted in one or more NIC task ratings. **Note:** An evaluation that resulted in

one or more NIC-rated tasks requires an investigation to determine the root cause of the identified deficiencies for each NIC task, and initiation of the decertification process on those respective tasks. See AFI 36-2670, for more details.

A2.1.1.9. Develop an executive summary that contains a summary of what was evaluated, the overall result (satisfactory or unsatisfactory), a comment about the technician's performance, strengths and weaknesses, and a deficiency trend analysis to include root causes and recommended management actions to prevent recurrence.

Table A2.1. Personnel Evaluation Task Deficiency Code.

CODE NUMBER	DESCRIPTION	CATEGORY ASSIGNMENT
P001	Tools or support equipment overdue inspection or calibration and utilized and/or tools or support equipment missing needed parts but used.	I or II
P002	Tools or support equipment damaged by improper use or handling.	I or II
P003	Applicable technical data not on hand, incorrect, out of date or not used.	I or II
P004	Technician not able to perform emergency procedures and/or did not comply with warnings, cautions, and notes.	I or II
P005	Technician lacked sufficient knowledge of the task and/or technician lacked sufficient knowledge of the operation of required support equipment.	I or II
P006	Controlling agencies not advised of changes in mission/equipment status when status changes due to task performance.	I or II
P007	Not all required steps performed (not the entire overall process) or not all steps were performed in the required sequence (not the entire overall process).	I or II
P008	An out-of-tolerance condition or an error in measurement was not recognized and resulted in the equipment not meeting technical data specifications.	I or II
P009	Inability/weakness in explaining weapon safety standards or nuclear related two-person concept requirements.	I or II
P010	Assets/equipment or materials not properly identified for turn in or disposal (this statement includes improper disposal of materials), and/or parts required to fix/clear equipment deficiencies not identified or ordered.	I, II, or III
P011	Not Otherwise Coded Deficiency.	I, II, or III
P012	Tools or support equipment improperly used or handled and/or repairable assets not properly identified for turn-in or repair.	II or III

P013	CFP not notified of changes in equipment status as a result of task performance and/or maintenance documentation not properly completed	II or III
P014	Lack of coordination with required agencies to ensure a safe, timely, and effective evaluation, and/or work area cleanup actions not accomplished.	II or III
P015	Tools or support equipment overdue inspection or calibration but not used and/or tools or support equipment missing needed parts but not used.	III
P016	Tools or support equipment not obtained before task initiation and/or tools or support equipment not properly stored after task completion.	III
P017	Technician not knowledgeable with emergency procedures and/or safety briefing not conducted when required and/or technician lacked sufficient knowledge of the operation of required support equipment documentation (e.g., sign out, sign off).	III
P018	Equipment status not checked to determine the existing condition and/or a JCN was not obtained for required documentation and/or no method available to document deficiencies discovered during the task performance.	III

Table A2.2. Personnel Evaluation Task Result Matrix.

Rule	If the Individual Committed	And	And	The result is
1	No Category I Deficiency	No Category II deficiency	No Category III deficiency.	C
2	No Category I Deficiency	Two or less Category II deficiencies	The accumulation of Cat III deficiencies did not detract from overall satisfactory job performance.	CWC
3	No Category I Deficiency	Two or less Category II deficiencies	The accumulation of Cat III deficiencies caused unsatisfactory job performance.	NIC
4	No Category I Deficiency	Three or more Category II deficiencies	N/A	NIC
5	One or more Category I deficiencies	N/A	N/A	NIC

A2.2. Report Submission.

A2.2.1. Evaluators will document PEs on AFMQCC 100-1. **(T-3).** Comments and recommendations are made on the evaluation report to eliminate the need for separate correspondence. The report(s) also provide a source for analyzing the effectiveness of the overall training program.

A2.2.1.1. Identify task deficiency codes and results, provide additional comments and references to each deficiency code, provide recommendations, and explain rescheduling actions, if required. Explain management, system or equipment deficiencies not directly reflecting on job performance in a separate special equipment report.

A2.2.1.2. The unit commander must review all evaluation reports with an overall result of unsatisfactory. **(T-3).** All other reports are reviewed at commander's discretion.

A2.2.1.3. If an individual's UTC evaluation result is unsatisfactory, the unit commander determines whether or not the individual's SORTS/ART/Defense Readiness Reporting System (DRRS) status is changed.

A2.2.1.4. PE reports will be routed only once through the routing chain. **(T-3).** At a minimum, include the evaluated individual, individual's supervisor, work center supervisor, and Unit Training Manager (UTM). For reports with an overall result of unsatisfactory, also include the superintendent/flight commander and unit commander. Reports will be closed within 30 calendar days from the date of evaluation as indicated on the AFMQCC 100-1. **(T-3).** The closing authority for "Satisfactory" evaluations is the flight commander. Closing authority for "Unsatisfactory" evaluations is the unit commander, but this may not be delegated below the Director of Operations.

A2.2.2. PE date completion will be annotated in IMDS, using screen 340. **(T-3).** Supervisors will annotate PE date completion as a journal entry in TBA. **(T-3).**

A2.3. Equipment Evaluation Report Writing and Submission.

A2.3.1. On EE reports, evaluators will:

A2.3.1.1. Use AF IMT 2420, *Quality Control Inspection Summary*, to document EE discrepancies, at the Commander's option. **(T-3).** EE discrepancies may be prepared in a narrative style on bond paper, or equivalent automated product, and attached to the AF Form 2419, *Routing and Review of Quality Control Reports*. **(T-3).**

A2.3.1.2. Categorize discrepancies as major or minor. **(T-3). Note:** A major discrepancy is any deficiency that degrades mission, system, or equipment capability; creates an unsafe condition; or may cause a security violation. A minor discrepancy is any deficiency that does not degrade mission, system, or equipment capability; does not create an unsafe condition; and is not a potential security violation.

A2.3.1.3. When EEs show no discrepancies, document this fact on AF Form 2419 or an equivalent automated product, and file as a completed report. **(T-3).**

A2.3.1.4. Document special EEs the same as above. **(T-3).** Unit Commanders may direct a special EE at any time. An example of a special EE would be when a piece of equipment is overhauled. Do not document special technical evaluations that reveal no discrepancies on an AF Form 2419 if an acceptance or commissioning document, such as an AFTO Form

217, *Certificate of Maintenance Accomplished* or AFTO Form 747, *Cyberspace Infrastructure System Acceptance* is signed. **(T-3).**

A2.3.2. The Flight Commander will be the closing authority for EEs with only minor discrepancies or without any discrepancies. **(T-3).** The Unit Commander, or his/her designee not below the Director of Operations, will be the closing authority for EEs with one or more major. **(T-3).**

Attachment 3

SAMPLING PLAN

A3.1. Common Terms.

A3.1.1. Population. The number of like items assigned to a work center will be used as the population for EEs. **(T-3).**

A3.1.2. Like Items. Like items perform the same basic function; are of the same basic design; and are maintained using the same basic tests and alignments.

A3.1.3. Sample Size. Sample size is the number of items that need to be sampled.

A3.2. Using a Sampling Plan for Equipment Evaluations.

A3.2.1. Count the number of like items and find the corresponding block under the column titled population size in **Table A3.1**, Sample Sizes.

A3.2.2. The corresponding sample size in the sample size column is the minimum number of like items to be inspected. For example, the unit has 25 A-Widgets and 30 B-Widgets which are considered like equipment end items. The total is 55; therefore, the sample size is 19 items. Nine (9) A-Widgets and ten (10) B-Widgets would be inspected over the 12-month timeframe.

Table A3.1. Sample Sizes.

Population Size	Sample Size
1	1
2 through 5	2
6 through 10	3
11 through 15	4
16 through 25	6
26 through 50	13
51 through 75	19
76 through 100	20
101 through 150	30
151 through 200	40
201 through 250	50
251 through 300	60
301 through 350	70
351 through 500	100
501 through 1000	200
1001-plus	25 percent of total

A3.3. Random Sampling.

A3.3.1. For other large quantities of materials that need to be sampled, use a sampling size of 25 percent. This sampling rate can be useful to inspect any large population commodity, such as TO files, publication files, bench stock bins, etc. Rather than inspect 100 percent of a file/commodity, evaluation of an appropriate sample size normally provides a reliable indication of the condition of the total file.

A3.3.2. This is the minimum required and units can increase this sample size depending on the deficiencies noted. For example, if performing a self-assessment on training and the determination is there is 145 training records consisting of the following: 60 3-levels, 70 5-levels and 15 7-levels. Technicians can either sample 25 percent of the 145 total (approx. 36) or 25 percent of each subcategory so 15 3-levels, 18 5-levels, and 4 7-levels with a total of 37 records. The latter provides a more comprehensive review and ensures an accurate evaluation on skill levels.

A3.4. Use of a Random Number Table.

A3.4.1. To effectively evaluate a random sample of commodity, technicians can use a random number generator to identify which item in the commodity to evaluate.

A3.4.2. To determine the specific items or records to be evaluated, develop a way to randomly identify the items. For example: enter the list of items to be evaluated into an Excel spreadsheet. After determining how many need to be inspected by multiplying the total by 0.25, use a sequence generator to help provide the number of an item to be sampled. If the sample plan determines technicians have 16 items, four items need to be sampled, and the sequence generator responds with 3, 5, 13, and 15 or technicians pull the corresponding number from a hat. Those items corresponding to those numbers are to be sampled. They were randomly selected to be evaluated. There are random sequence generators available via the Internet.

Attachment 4

AIR FORCE MAINTENANCE QUALITY CONTROL CHECKLISTS

A4.1. AFMQCC's Currently Available. The AFMQCCs can be found at: https://usaf.dps.mil/sites/affsa/SitePages/XAR.aspx. **NOTE**: Those equipment titles with an "*" can be found at https://cs2.eis.af.mil/sites/12994/Documents/Forms/AFCQCC.aspx.

Table A4.1. AFMQCC Listing.

Number	Title
AFMQCC 100-1	Personnel Evaluation Report
AFMQCC 300-1	FRN-44A
AFMQCC 300-2	FRN-45C
AFMQCC 300-11	FMQ-19
AFMQCC 300-13	GRN-29v1
AFMQCC 300-14	FMQ-13
AFMQCC 300-15	FMQ-22
AFMQCC 300-16	FMQ-23
AFMQCC 400-2	GPN-22
AFMQCC 400-3	TPX-42
AFMQCC 400-8	FA-10221
AFMQCC 400-9	*TPS-75
AFMQCC 400-10	FA-10095
AFMQCC 400-11	GPN-20
AFMQCC 400-12/13	FSQ-204
AFMQCC 400-14/15	GPN-30
AFMQCC 400-16	MPN-14K
AFMQCC 500-1	GRT-21/22
AFMQCC 500-2	GRR-23/24
AFMQCC 500-3	OK-423
AFMQCC 500-4	GRC-211
AFMQCC 500-5	GRC-171
AFMQCC 500-11	FSC-127a
AFMQCC 500-13	GSH-74
AFMQCC 500-14	MSN-7
AFMQCC 1500-1	TRN-26
AFMQCC 1500-2	TRN-41
AFMQCC 1500-3	TRN-48

A4.2. Future AFMQCCs. Table A4.1. is subject to change as centrally managed equipment assets are added or deleted from the inventory.

Attachment 5

AIR FORCE MAINTENANCE SPECIAL INSTRUCTIONS

A5.1. AFMSIs Currently Available. The AFMSIs can be found at: https://usaf.dps.mil/sites/affsa/SitePages/XAR.aspx

Table A5.1. Air Force Maintenance Special Instruction Listing.

Number	Title
AFMSI 300-1	Scheduled periodic inspections, ATC Communications Systems
AFMSI 300-6	Scheduled periodic inspections, FSC-127

A5.2. Future AFMSIS. Table A5.1. is subject to change as AFMSIs are added or deleted.

Attachment 6

NEXT GENERATION RADAR OPERATIONS CENTER SUPPORT

A6.1. NEXRAD Operations. The DoD owns and maintains a total of 25 Weather Surveillance Radar-1988, Doppler (WSR-88D) systems. In accordance with a tri-agency agreement among the DoD, Commerce, and Transportation, the Radar Operations Center (ROC) located in Norman, Oklahoma is responsible for operational support to the WSR-88D network and can be contacted at 1-800-643-3363.

A6.2. ROC Support. ROC support to WSR-88D locations include:

A6.2.1. Centralized radar operations support, field assistance, software maintenance, engineering support, and special depot-level support of WSR-88D units.

A6.2.2. Systematic and coordinated analyses of the day-to-day operations and maintenance of WSR-88D units to determine the need for improvements and immediate/long-term life cycle support.

A6.2.3. Analyzing, developing, testing, and evaluating proposed changes to WSR-88D hardware/software configurations, materials, techniques, and procedures.

A6.2.4. Implementation of approved WSR-88D hardware, software, and documentation changes.

A6.2.5. Monitoring the FAA OE/AAA database for proposed and ongoing construction/wind turbine projects in the vicinity of WSR-88D locations.

A6.2.6. Assessments of proposed and ongoing construction/wind turbine projects in the vicinity of WSR-88D locations to determine potential impacts to weather radar operations.

A6.2.7. Notifying the DoD Focal Point when proposed and ongoing construction/wind turbine projects threaten to affect WSR-88D locations and their capability to provide weather support to DoD installations and missions.

A6.2.8. Assisting the Air Staff EMWG, MRT, and affected MAJCOM/FOA/DRU weather functional, with efforts to mitigate potential impacts to WSR-88D operations from proposed and ongoing construction/wind farm projects in accordance with AFI 90-2001.

Attachment 7

CRITICAL OUTAGE REPORTING

A7.1. Critical Outages. A COR will only be used to report systems that are NMC as defined in AFI 21-103. **(T-3).**

Table A7.1. Critical Outage Report Template.

Critical Outage Report
1. UNIT/BASE: Self-explanatory.
2. TYPE REPORT: (Initial, Update, Closing).
- Initial: First report during a NMC (Red) outage
- Update: Report providing significant details regarding change in status
- Closing: Last report indicating outage is closed and system is Fully Mission Capable (FMC) (Green)
3. AFFECTED SYSTEM: Equipment designator and Nomenclature (e.g. AN/GPN-30, DASR).
4. JOB CONTROL NUMBER: JCN assigned and loaded into IMDS.
5. DATE/TIME OUT: Start NMC time (Local).
6. DATE/TIME IN: Start Partially Mission Capable (PMC)/FMC time (Local).
7. ESTIMATED TIME TO REPAIR: Anticipated PMC/FMC time.
8. JOB STATUS: AWP, Mission In Progress (MIP), Weather Delay, Single Shift Delay, etc.
9. NARRATIVE DESCRIBING OUTAGE: Cause, Condition, Symptoms of outage.
10. NARRATIVE DESCRIBING ACTIONS TAKEN: Detailed synopsis of maintenance actions/results.
11. NAME AND STOCK NUMBER OF ITEM / PARTS ORDERED: Self-explanatory.
12. DOCUMENT NUMBER: Info for item(s) listed in Line 11.
13. OFF BASE REQUISITION NUMBER / PRIORITY: Info for item(s) listed in Line 11
(FBXXXXXXXXX).
14. POC/RELEASING OFFICIAL: Name, rank, and telephone number of commander/superintendent or NCOIC responsible for the production work center.
15. MISSION IMPACT: Detailed synopsis of the outage's impact on base flight operations. Include name, rank, office symbol, and telephone number of person providing mission impact.

Table A7.2. Critical Outage Report Equipment List.

ASR-9
DBRITE
FPN-68A
FRN-43, 44, 45 (all versions)
FSC-127 (>= 50% Failure)
FSQ-204
GPN-22
GPN-30
GRN-29 (all versions, all subsystems)
GSH-74
MPN-25
MSN-7
OK-235
OK-236
Outage affecting entire ATCALS receiver/transmitter/GATR site
TPN-19
TPN-24
TPN-25
TRN-28
TRN-41
UPX-37

Table A7.2. (AETC) Critical Outage Report Equipment List.

Note: This list should include any full weather suite version, as well as any additional RAWS, maintained equipment, or system identified by the RAWS MAJCOM Functional Manager (MFM).